Can molecular-dynamics DNA shape alone predict Gata4 HT-SELEX affinities? New tetramer/hexamer MD tables, preprocessing, model comparison and LRP

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23 October 2026

Abstract

Aim. We ask whether good prediction of protein–DNA affinity is possible from molecular-dynamics (MD) simulation data alone, using newly generated tetramer and hexamer tables of average helical parameters (`avg`) and stiffness matrices (`fce`) to predict Gata4 HT-SELEX 10-mer affinities. **Data and protocol.** Labels come from the SELEX affinity table (`4.txt`), in which affinity is exactly $0.172\,(\text{obs/exp})^{1/4}$, only 12.1% of 10-mers are present and 52.5% of entries have a single count. A preprocessing ablation under a leakage-free scheme (5-fold CV grouping each k-mer with its reverse complement, plus a fixed 10% held-out test set) selected log-affinity labels, a count threshold of $\geq$10 and per-strand labels (gain of 0.15–0.17 CV R^2 over merged labels). Eight regressors (Linear, Ridge, SVR, random forest, DFN, CNN1D, CNNAttention, Transformer) were benchmarked on seven feature sets: tetramer, hexamer or tetramer+hexamer MD shape, each with one-hot (a–c) and *without any one-hot* (e–g), and one-hot only (d). **Results.** MD-only prediction works: with tetramer MD only, CNN1D reaches a held-out test $R^2 = 0.700 \pm 0.009$ (Pearson 0.838, Spearman 0.821; 0.751 averaged over its 15 runs), against 0.739 for one-hot only and 0.736 for MD + one-hot, i.e. 95% of the sequence-based value. CNNAttention (0.683), the Transformer (0.649), DFN (0.606), RF (0.562), SVR (0.547) and Linear/Ridge (0.416) follow. For the linear models and RF, MD features beat one-hot (+0.036 and +0.027); for the other models MD-only is 0.03–0.05 below one-hot. Hexamer MD alone is worst (CNN1D 0.553). The MD tables are tetramer lookups that identify the sequence (central bases decoded linearly with 100% accuracy, outer bases 76%). A control that randomly reassigns the MD rows to tetramers shows that the real values are worth +0.135 R^2 for DFN, +0.052 for RF and +0.022 for CNN1D, so most of the CNN's MD-only accuracy (0.678 of 0.700) is attainable with any identity code. Layer-wise relevance propagation shows that the MD-only models rely mostly on `fce` coupling terms (57–64%), with TILT–ROLL, SLIDE–TILT and ROLL–TWIST among the largest, and RISE the clearest positive effect on strong binders. The strongest binders carry an ETS-like ACTTCCGG motif, not GATA; `4.txt` shows no GATA enrichment. The supplied `SELEX_test.txt` is not consistent with `4.txt` and was not used. **Verdict.** MD data alone predict affinity well, but the evidence for physical rather than identity-code use is strong only for models that cannot learn their own code (DFN, RF, linear).

1 Introduction

Transcription factors (TFs) recognise their genomic targets through direct base readout and through the sequence-dependent three-dimensional shape of DNA ("shape readout") [5]. High-throughput SELEX (HT-SELEX) enriches DNA ligands bound by a TF over successive selection rounds and sequences them, giving large-scale relative affinities for k-mers [1]. Physics-based descriptors from atomistic molecular dynamics (MD) summarise the local structure and flexibility of DNA. Examples are average base-step helical parameters (shift, slide, rise, tilt, roll, twist) and stiffness (force-constant) matrices. Such descriptors can complement or replace the raw sequence

as model input, and they give an interpretable view of affinity determinants. DNAffinity builds regressors on such features. Here we test new tetramer- and hexamer-level MD shape tables on Gata4 HT-SELEX data.

Aim. The central question is whether *good affinity prediction can be achieved from MD simulation data only.* MD-derived shape and flexibility are physics-based features: they encode what atomistic simulation says about how a DNA sequence bends, twists and fluctuates, without any explicit sequence input. If they predict affinity as well as the sequence itself, they could provide an interpretable, transferable description of DNA recognition. A caveat is that every MD table is a lookup indexed by k-mer sequence, so MD features can in principle simply act as a sequence identifier. We therefore ask: (i) how well do MD-only features (sets e–g) predict compared with one-hot only (d) and MD + one-hot (a–c); (ii) how much sequence information is recoverable from the MD features; (iii) whether models use the tables as identity codes or as physical descriptors (permuted-table control); and (iv) which MD parameters drive the predictions (LRP). Secondary questions are which preprocessing is optimal and how classical and deep models compare.

2 Methods

2.1 Affinity labels

Labels are taken from the SELEX package output for Gata4 (`data/selex/Gata4/4.txt`). It has 126,624 10-mers with observed and expected (Markov-model) counts, affinity and SE. We verified that Affinity $= 0.172\,(\text{obs}/\text{exp})^{0.25}$ to machine precision (sd of the ratio 2.5×10^{-16}) and that $\text{SE}\sqrt{n}/\text{Aff}$ has median $1.414 \approx \sqrt{2}$. The table is zero-truncated: only 12.1% of the 4^{10} possible 10-mers appear. 52.5% of entries have count 1, and 13,100 have count $\geq$10. Because the SE is about $\sqrt{2}\,\text{Aff}/\sqrt{n}$, labels with low counts are very noisy.

2.2 Shape features

The new tables `data/selex/tetramer/{avg,fce}_tetramer.dat` and `data/selex/hexamer/{avg,fce}_hexamer.dat` give, for every tetramer (256) and hexamer (4096), six average helical parameters (`avg`) and a flattened 6×6 stiffness matrix (`fce`). The matrices are exactly symmetric and positive definite (minimum eigenvalue $\approx$ 0.006), so we keep the 21 unique upper-triangle entries. A 10-mer is represented by 7 overlapping tetramer windows (and 5 hexamer windows). Hexamer j is aligned to its central tetramer window j+1 and zero-padded at the ends for windowed models. Optional one-hot input adds the 16 one-hot values of the bases in each tetramer window. Flat models use the concatenated windows; windowed models (CNN1D, CNNAttention, Transformer) use a $7 \times d$ tensor.

2.3 Data processing pipeline

All steps are implemented in `scripts/selex_new_shape/lib.py` (functions `load_labels`, `split`, `flat_feats`, `windowed`, `tf`) and `train_final.py`. Figure 1 summarises them.

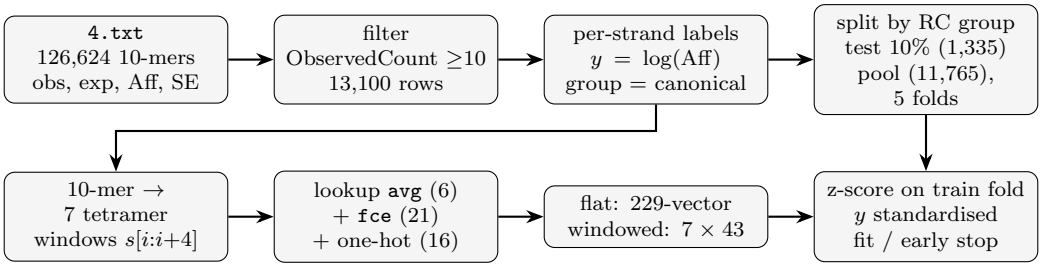


Figure 1: Processing pipeline for the chosen configuration. Row counts are those of the final runs.

2.3.1 Reading and checking the labels

`4.txt` (tab-separated, quoted header, R row names) is read with pandas. The columns used are `Kmer`, `ObservedCount` (n), `ExpectedCount` (e, from the Markov background model of round 0) and `Affinity`. We checked that

$$\mathrm{Aff} = 0.172\,(n/e)^{1/4}, \qquad \mathrm{SE} \approx \sqrt{2}\,\mathrm{Aff}/\sqrt{n}, \tag{1}$$

so $\mathrm{SE}/\mathrm{Aff} = \sqrt{2/n}$: 141% at $n = 1$ and 45% at $n = 10$. The table is zero-truncated. Only k-mers seen at least once in the selected round are listed (12.1% of 4^{10}), and there are no entries for unobserved, low-affinity k-mers. The models therefore learn on the observed range only.

2.3.2 Count filter

Rows with $n \geq 10$ are kept (`load_labels(min_count=10)`), leaving 13,100 of 126,624 rows. This bounds the relative SE at about 45%. The ablation (Table 2) chose this threshold.

2.3.3 Per-strand vs. merged labels

Each row's group key is the canonical sequence $\min(s, \mathrm{rc}(s))$, with `canonical` from `dnaffinity.preprocess`.

- *Per-strand* mode (chosen) keeps each measured orientation as its own row with its own label. A reverse-complement row exists only if that orientation itself passed the filter. The 13,100 rows form 10,771 groups.
- *Merged* mode averages the affinities of the measured orientations per group and emits the canonical sequence and its reverse complement with the same label (reverse-complement augmentation).

2.3.4 Label transform

The target is $y = \log(\mathrm{Aff}) = \frac{1}{4}\log(n/e) + \log 0.172$, i.e. a scaled log-enrichment. Inside each training fold, y is also standardised for model fitting, $\tilde{y} = (y - \bar{y}_{\mathrm{tr}})/s_{\mathrm{tr}}$, and predictions are transformed back. All metrics are computed on y.

2.3.5 Windowing and shape lookup

A 10-mer s is cut into the 7 overlapping tetramers $s[i{:}i{+}4]$, $i = 0, \ldots, 6$ (each describes the central base step of its window). For each window we look up:

- the 6 averages of `avg_tetramer.dat` (shift, slide, rise, tilt, roll, twist);
- the 21 upper-triangle entries (including the diagonal) of the symmetric 6×6 stiffness matrix in `fce_tetramer.dat`. The 36 stored values contain 15 duplicates.

Hexamer tables, used only in the ablation, are read the same way. A flat vector concatenates the 5 hexamers $s[j{:}j{+}6]$. In windowed form, hexamer j is placed at tetramer position $j{+}1$, i.e. aligned on its central tetramer, and the two end positions are zero-padded.

2.3.6 One-hot encoding

What it does. One-hot encoding turns each base into a 4-dimensional binary vector with a single 1 at the position of that base in the fixed order (A, C, G, T): A $\to (1,0,0,0)$, C $\to (0,1,0,0)$, G $\to (0,0,1,0)$, T $\to (0,0,0,1)$ (`lib.OH`).

Why it is needed. The shape tables describe each tetramer only indirectly, through 27 physical numbers. One-hot gives the models the sequence identity itself. Coding bases as integers (e.g.

A=0, …, T=3) would impose a false order and spacing (T "larger" than A, G "between" C and T). One-hot has no such ordinal bias: all four bases are equidistant, and every weight that multiplies a channel is specific to one base at one position, like a position weight matrix.

Layout in `lib.py`.

- **Flat models** (`flat_feats`): the 10 bases of the 10-mer are encoded in order and concatenated, giving $10 \times 4 = 40$ values. They come after the shape block: [`avg` (42), `fce` (147), one-hot (40)] for set (a). Channel $4j + n$ of the one-hot block is 1 if base j ($j = 0..9$) is nucleotide n.
- **Windowed models** (`windowed`): at tetramer window i ($i = 0..6$, bases $s[i{:}i{+}4]$) the 4 bases of the window are encoded and concatenated, giving 16 values per position. They are appended after the 27 shape channels of the same window, so each position has [`avg` (6), `fce` (21), one-hot (16)] = 43 channels. Because consecutive windows overlap by 3 bases, every internal base appears in up to 4 windows, at a different offset in each.
- **One-hot only** (set (d), `shape=0`): the shape block is omitted, giving 40 flat values or 7×16.

Worked example. For the tetramer window GATA:

$$\mathrm{G}\,(0,0,1,0) \parallel \mathrm{A}\,(1,0,0,0) \parallel \mathrm{T}\,(0,0,0,1) \parallel \mathrm{A}\,(1,0,0,0) \;=\; (0,0,1,0,\ 1,0,0,0,\ 0,0,0,1,\ 1,0,0,0).$$

In a windowed model these 16 values are concatenated with the 6 `avg` values (SHIFT, SLIDE, RISE, TILT, ROLL, TWIST) and the 21 `fce` values that the shape tables give for GATA.

Combination with shape. Shape and one-hot are concatenated, never multiplied. Like all features, the one-hot channels are then z-scored with the training-fold mean and sd. A channel $x \in \{0, 1\}$ with frequency p becomes $(x-p)/\sqrt{p(1-p)}$, so an absent base gets a non-zero (negative) value. The shape features are a deterministic function of the tetramer and hence of its 16 one-hot values. A model can therefore read sequence identity directly (one-hot) and/or use the physical summary (shape). This redundancy is what the one-hot-only benchmark (d) and the LRP analysis test.

2.3.7 Feature tensors per model

- **Flat** (Linear, Ridge, SVR, RF, DFN): [$\texttt{avg}_{0..6}$ (42), $\texttt{fce}_{0..6}$ (147), one-hot (40)], i.e. a 229-dimensional vector (`flat_feats`).
- **Windowed** (CNN1D, CNNAttention, Transformer): a 7×43 tensor whose channels per position are [`avg` (6), `fce` (21), one-hot (16)] (`windowed`).

2.3.8 Standardisation

Within each CV fold, features are z-scored with means and standard deviations computed on the training-fold rows only and applied to all rows (validation and test). For windowed input the statistics are pooled over samples and positions per channel. Because of this external scaling, the input `LayerNorm` of the PyTorch models was switched off (`normalization=False`).

2.3.9 Splits

`split()` draws 10% of all canonical groups in `4.txt` (seed 20) as the test set. This gives 1,335 test rows and 11,765 pool rows after filtering, the same set for every model and configuration. The pool is partitioned by `GroupKFold(5)` on the canonical group, so a k-mer and its reverse complement are always on the same side. For NNs, a further random 10% of each training fold (seed $100 + k$ for fold k) is the early-stopping set, and reported "train" metrics include it.

2.4 Feature sets benchmarked

Seven feature sets were compared. *Sets (a)–(c) are shape + one-hot*: the MD-derived shape windows from `data/selex` (`avg`+`fce`) concatenated with the one-hot encoding. Set (d) is *one-hot only*, with no shape. *Sets (e)–(g) are MD shape only* (`onehot=0`, `shape=1`): the same shape windows as (a)–(c) but without any one-hot channel, so the model sees the sequence only through the MD tables. Sets (d)–(g) were run with the full final protocol (outputs `final_onehot`, `final_e`, `final_f`, `final_g`), not taken from the ablation. Everything else was identical across sets: labels from `4.txt` (the shape files have none), log affinity, count $\geq$10, per-strand rows, per-fold standardisation, and the same test split and grouped folds. Per window, `avg` has 6 values and `fce` 21.

Table 1: Input dimensions per feature set. Flat models: Linear, Ridge, SVR, RF, DFN. Per-position models: CNN1D, CNNAttention, Transformer. One-hot is 40 values (flat) or 16 per position. †Hexamer channels are zero at positions 0 and 6.

Feature set	Flat vector	Per-position tensor
(a) tetramer shape + one-hot	$7 \times 27 + 40 = 229$	$7 \times (27 + 16) = 7 \times 43$
(b) hexamer shape + one-hot	$5 \times 27 + 40 = 175$	$7 \times (27 + 16) = 7 \times 43^{\dagger}$
(c) tetra+hexa shape + one-hot	$7 \times 27 + 5 \times 27 + 40 = 364$	$7 \times (27 + 27 + 16) = 7 \times 70$
(d) one-hot only	40 (10×4)	7×16 (bases of each tetramer window)
(e) tetramer shape only	$7 \times 27 = 189$	7×27
(f) hexamer shape only	$5 \times 27 = 135$	$7 \times 27^{\dagger}$
(g) tetra+hexa shape only	$7 \times 27 + 5 \times 27 = 324$	$7 \times (27 + 27) = 7 \times 54$

In the per-position layout, hexamer $s[j{:}j{+}6]$ ($j = 0..4$) is placed at tetramer position $j{+}1$, its central tetramer. The two end positions get zero hexamer channels, which become constant values after standardisation (`lib.windowed`). Flat models concatenate the 5 hexamer windows without padding (`lib.flat_feats`). Set (a) is the chosen preprocessing; its runs are the final runs of Section 2.9.

2.5 Timing

All timings were measured on the box: 8-core Intel Xeon CPU (1 thread per core), 15 GB RAM, no GPU, PyTorch on CPU.

- Jobs ran as 8 parallel workers. NNs, Linear, Ridge and SVR used 1 thread each; RF used 6 threads (`n_jobs`=6). Workers competed for cores, so times are indicative and comparable within this benchmark.
- *Feature time* is building the feature matrix for all 13,100 rows.
- *Training time* per run is the wall time of `fit` (including early stopping) plus one prediction pass over all rows.
- *Inference* is a separate `predict` on the 1,335 test rows, scaled to 1,000 sequences.
- The original tetramer runs did not log timing. They were re-timed by rerunning the same code with seed 0 for every fold (5 runs per model) under the same parallel setup. Their reported total is the per-run mean times the number of runs.

2.6 Is the MD signal physical or an identity code? Two controls

The MD tables are lookups: every tetramer has its own 27 numbers, so shape-only input can in principle just label the tetramer, like a one-hot code. We therefore ran two controls.

- **Identity check** (`identity_check.py`). For the 256 tetramers (z-scored 27-dimensional vectors) we report the rank, the smallest pairwise distance, and the cross-validated accuracy (grouped by reverse complement) of a logistic regression that predicts the base at each tetramer position from the 27 values. We also test whether all tetramers are still recovered by nearest neighbour after adding Gaussian noise (sd 0.1 in z-units).
- **Permuted-table control.** The 256 real MD rows (`avg` and `fce` together) are randomly reassigned to the tetramers (`PERM_TAB`; seeds 1 and 2). The input then has identical dimension, value distribution and identifying power, but the MD values no longer correspond to the physical properties of the tetramer. If real MD values only act as an identity code, the permuted table should perform the same. The control is run for set (e) with RF (5 runs), DFN and CNN1D (15 runs each), with the same protocol, folds and seeds. It was not run for the other models or sets.

2.7 Validation protocol

Every k-mer is grouped with its reverse complement. A fixed random 10% of groups (seed 20, drawn over all groups in `4.txt`) forms the held-out test set. The remaining pool is split by `GroupKFold` into 5 folds. Inside each training fold, neural networks hold out a random 10% for early stopping. For every fold model we report train-fold, validation-fold and test metrics. Metrics are R^2, Pearson and Spearman correlations on log affinity, given as mean±sd over folds (and seeds), plus the train–test R^2 gap. The "averaged-runs" test R^2 is the R^2 of the mean prediction of all fold/seed models of one type.

2.8 Preprocessing ablation

Starting from a base configuration (log label, count $\geq$10, per-strand labels, tetramer avg+fce, standardisation), we changed one factor at a time and scored a random forest (300 trees, min leaf 3, max features 0.33) and a CNN1D (one seed). The factors were: label transform (raw vs log), count threshold (1, 3, 10, 30), merged vs per-strand labels, feature tables (tetramer, hexamer, both), `avg`/`fce`/both, one-hot ± shape, and standardisation. Merged labels average the measured strands per canonical k-mer and emit both orientations (reverse-complement augmentation). Per-strand labels keep each measured orientation as its own row. To make configurations comparable, every one was scored on the same rows: held-out-fold rows with a measured per-strand label and count $\geq$10. The CNN was not run at thresholds 1 and 3 because of CPU cost.

2.9 Models

The repository models are used through their `dnaffinity.models` classes. The architectures below are taken from that code (repository `PieterZanders/DNAffinity`, branch `deep-learning`). Parameter counts are for our 229-feature / 7×43 inputs.

Shared PyTorch training loop (`torch_base.TorchRegressor`).

- AdamW optimiser, MSE loss (Huber optional) and gradient-norm clipping at 5.
- ReduceLROnPlateau (factor 0.5, patience $\lfloor$patience/3$\rfloor$).
- Early stopping on the validation loss, with the best weights restored.
- Shuffled mini-batches; batches of size 1 are skipped because of BatchNorm.
- Our settings: batch size 64, at most 300 epochs, patience 25, weight decay 10^{-2}, 3 seeds (0, 1, 2) per fold.

2.9.1 Linear regression

`dnaffinity.models.Linear` wraps scikit-learn `LinearRegression` (ordinary least squares, no penalty).

- *Pros:* convex, deterministic and fast. The coefficients can be read directly.
- *Cons:* it cannot represent interactions or non-linear dependence on shape.
- *Biological relevance:* on the one-hot block it is an additive, position-independent model, equivalent to a position weight matrix (PWM) on log affinity. On the shape block it is an additive model of base-step geometry and stiffness.
- *Observed:* it underfits (train R^2 0.50, test 0.47, gap 0.03). Additivity explains about half of the variance of log affinity.

2.9.2 Ridge regression

`dnaffinity.models.Ridge` wraps scikit-learn `Ridge` (L2 penalty). The repository default is $\alpha = 1$; we used $\alpha = 10$.

- *Pros:* the penalty stabilises the strongly collinear shape features (overlapping windows, correlated parameters).
- *Cons:* same additive limit as linear regression.
- *Observed:* practically identical to linear regression (test 0.470). With 11.7k rows and 229 features the variance of OLS is already small, so the limitation is bias, not variance.

2.9.3 Support vector regression

`dnaffinity.models.SVR` wraps scikit-learn `SVR` with the repository defaults: RBF kernel, $C = 1$, $\varepsilon = 0.1$, $\gamma =$ 'scale', on standardised inputs and targets.

- *Pros:* a smooth non-linear kernel model that captures interactions implicitly and is robust thanks to the ε-insensitive loss.
- *Cons:* cost grows between quadratically and cubically with the number of samples. It is not interpretable, and the isotropic RBF kernel treats all 229 features alike.
- *Biological relevance:* similarity in kernel space means sequences with similar shape and base profiles get similar affinity, a nearest-neighbour view of the binding landscape.
- *Observed:* test 0.598, gap 0.146, better than RF.

2.9.4 Random forest

`dnaffinity.models.RandomForest` wraps scikit-learn `RandomForestRegressor` (repository defaults: 200 trees, `max_features`=1.0). We used 500 trees, `min_samples_leaf`=3 and `max_features`=0.33.

- *Pros:* non-parametric and indifferent to feature scale. It gives feature importances that can rank shape parameters per window.
- *Cons:* axis-aligned splits approximate smooth effects poorly. Deep trees memorise noisy labels.
- *Biological relevance:* threshold rules on single shape parameters (e.g. "roll at step 4 above x") are directly readable as indirect-readout hypotheses [5].

- *Observed:* largest gap (train 0.884, test 0.572). It is also the model that benefited most from shape on top of one-hot in the ablation (0.563 → 0.599), because the shape tables give trees ready-made smooth physical covariates.

2.9.5 DFN (deep feedforward network)

`dfn.DFN` builds `FeedForward`: `nn.Linear` layers with `ReLU` and `Dropout` after every hidden layer and no activation on the output. It is wrapped in `RegressionCV`, which adds an optional input `LayerNorm` (disabled here). The repository default layers are `[64,32,1]` (and `[256,128,64,32,1]` with lr 10^{-4} for SELEX in `common.py`). We used `[256,128,64,1]` (229→256→128→64→1, 100,097 parameters), dropout 0.2 and lr 3×10^{-4}.

- *Pros:* a universal approximator on the flat vector that can learn any interaction.
- *Cons:* no positional inductive bias (a motif shifted by one window is a new pattern), so it needs more data and overfits easily.
- *Observed:* test 0.644, gap 0.176, best epoch 44 of 70, the best flat model.

2.9.6 CNN1D

`cnn1d.CNN1DRegressor` builds `CNN1D` (60,481 parameters):

- optional input `LayerNorm` (off);
- stem `Conv1d`(43→64, kernel 3, padding 1) + `BatchNorm1d` + `ReLU`;
- 2 `ResidualConvBlocks`, each Conv1d(64,64,3)–BN–ReLU–Conv1d(64,64,3)–BN, a skip connection, then ReLU;
- global average pooling over the 7 positions;
- `MLPHead`: Linear(64→32)–ReLU–Dropout–Linear(32→1).

We used the repository defaults (hidden 64, 2 blocks, kernel 3, average pooling) with dropout 0.2 and lr 10^{-3}.

- *Pros:* weight sharing gives translation equivariance, so the same filter detects a feature at every position. Capacity is moderate.
- *Cons:* the receptive field is limited by depth (here the 2 blocks plus stem cover the whole 7-window input). Pooling discards absolute position.
- *Biological relevance:* first-layer filters over one-hot channels act like PWMs/motif detectors, and stacked convolutions combine adjacent sub-motifs. Pooling encodes the idea that a site can sit at any offset in the 10-mer, which matters here because the top k-mers are shifted overlaps of the same motif.
- *Observed:* best single model (test 0.736, 0.788 for averaged runs). It overfits after epoch ≈27 (final train MSE 0.068 vs. best validation 0.261), which early stopping controls. Shape channels were redundant given one-hot: 7 windows with 16 one-hot channels already determine the tetramer, so the network can learn any per-tetramer lookup itself.

2.9.7 CNNAttention

`cnn_attention.CNNAttentionRegressor` uses the same `ConvBackbone` as CNN1D, followed by `num_attn_layers`=1 `nn.MultiheadAttention` layer (embedding 64, 4 heads, dropout as configured) with a residual connection and `LayerNorm`, then global average pooling and the same `MLPHead` (77,249 parameters; repository defaults otherwise).

- *Pros:* attention can couple distant positions, beyond the convolutional receptive field, and its weights can be inspected.
- *Cons:* more parameters, and little extra range on a length-7 input.
- *Biological relevance:* pairwise attention can represent dependencies between non-adjacent positions (dinucleotide/epistatic effects, spacing of half-sites), which a PWM ignores.
- *Observed:* test 0.725, gap 0.155, slightly below CNN1D. For 10-mers the convolutions already see the whole sequence, so attention adds capacity but no new range.

2.9.8 Transformer

`transformer.TransformerRegressor` builds the following (19,073 parameters):

- optional input `LayerNorm` (off) and a `Linear` projection $43 \to d_{\text{model}}$;
- sinusoidal positional encoding and a learnable CLS token prepended;
- `nn.TransformerEncoder` with pre-norm layers (`norm_first=True`, GELU activation);
- pooling over the CLS token or the mean of the position tokens;
- an `MLPHead` as above.

We used the `common.py` defaults for this model: $d_{\text{model}} = 32$, 2 layers, 4 heads, feed-forward 64, mean pooling, lr 3×10^{-4} with 5 warm-up epochs. The code comment notes that CLS pooling at lr 10^{-3} collapsed on SELEX. We used dropout 0.2.

- *Pros:* global, content-dependent interactions between all positions, with explicit position information.
- *Cons:* weak locality bias (no built-in notion that adjacent bases matter most). It needs much data and careful optimisation, and was the slowest to train here.
- *Biological relevance:* like attention above, it can model position-specific coupling and orientation-dependent context. Explicit positional encoding can in principle use the absolute-position and strand effects behind the label asymmetry.
- *Observed:* test 0.685, smallest gap (0.077), train MSE about equal to validation MSE (0.293 vs. 0.291), best epoch ≈112. It underfits relative to the CNNs at this size, consistent with its weaker inductive bias on about 10k training rows.

2.9.9 Averaged runs

For each model type, the "averaged-runs" prediction is the mean of the test-set predictions of all its fold×seed models (5 for non-NN models, 15 for NNs). Models of different types are never combined.

- *Pros:* averaging reduces the variance from seeds and data subsets. This is the main reason CNN1D rises from 0.736 to 0.788.
- *Cons:* the training cost is multiplied by the number of runs, and so is the inference cost when deployed.

2.9.10 Shape features and biophysics

The `avg` and `fce` channels encode indirect readout: the intrinsic geometry and deformability of each base step, which affect the energetic cost of the DNA conformation a TF binds [5]. They help models without a positional, sequence-aware inductive bias (RF: +0.036 over one-hot alone). For CNN1D they are a deterministic function of the one-hot input and gave no gain. All shape tables are reverse-complement symmetric (shift and tilt change sign), so they cannot by themselves explain the observed strand asymmetry of the labels. Only orientation-aware inputs (per-strand rows, position-specific channels) can represent it.

2.10 Feature importance with Layer-wise Relevance Propagation

Theory. LRP [2, 3] redistributes a model output $f(x)$ backwards, layer by layer, onto the input features. Each neuron k passes its relevance R_k to its inputs j in proportion to their contributions $z_{jk} = a_j w_{jk}$:

$$R_j = \sum_k \frac{a_j \, \rho(w_{jk})}{\epsilon \, \mathrm{sign}(z_k) + \sum_{j'} a_{j'} \rho(w_{j'k}) + \rho(b_k)} \, R_k.$$

The plain rule (LRP-0) uses $\rho(w) = w$ and $\epsilon = 0$. LRP-ϵ adds a stabiliser ϵ. LRP-γ uses $\rho(w) = w + \gamma w^+$, which favours positive contributions and gives less noisy maps in deep conv layers. When biases are zero and $\epsilon \to 0$, relevance is conserved: $\sum_j R_j = f(x)$. Relevance assigned to biases and to ϵ is lost, so conservation is only approximate. Relevance is signed: positive relevance pushes the prediction up, negative pushes it down.

Implementation and rules. We wrote a manual LRP for the exact repository architectures (`scripts/selex_new_shape/lrp.py`). Every layer is covered explicitly; Captum or zennit was not needed.

- **DFN** (`Linear`, `ReLU`, `Dropout`): all `Linear` layers use LRP-ϵ ($\epsilon = 0.01$). ReLU passes relevance unchanged, and Dropout is the identity in evaluation mode.
- **CNN1D** (`Conv1d`, `BatchNorm1d`, `ReLU`, residual addition, global average pooling, MLP head):
  - Each `BatchNorm1d` is folded into its preceding `Conv1d`, giving $w' = w\,\gamma_{\mathrm{bn}}/\sigma$ and $b' = (b - \mu)\gamma_{\mathrm{bn}}/\sigma + \beta$. This is exact in evaluation mode; the folded network reproduces the predictions to 10^{-4}, which is asserted in the code.
  - The residual addition $o + h$ and the global average pool are linear, so relevance is split in proportion to their summands (ϵ-rule).
  - The hidden convolutions inside the residual blocks use LRP-γ ($\gamma = 0.25$). Their inputs are post-ReLU and therefore non-negative.
  - The input convolution uses LRP-ϵ. The inputs are z-scored and can be negative, so LRP-γ and z^+ rules are not applicable there.
  - The head `Linear` layers use LRP-ϵ.
- No layer was unsupported. The optional input `LayerNorm` is disabled in our models (`normalization=False`). If it were enabled, it would need a dedicated rule, e.g. treating the normalisation as constant.

Relevance explains the model output in standardised log-affinity units ($f(x)$ before de-standardisation), relative to an all-zero input. The all-zero input corresponds to the training-fold mean feature vector.

Checks.

- **Correctness.** For piecewise-linear networks, LRP-0 equals gradient×input [4]. We therefore also ran LRP-0 and compared it with autograd gradient×input.
- **Conservation.** We report, over test rows, the correlation and slope between $\sum_j R_j$ and $f(x)$, and the median ratio $\sum_j R_j / f(x)$ for $|f(x)| > 0.5$.

Weights and data. The final protocol did not save weights, so DFN and CNN1D were retrained with the identical code, configuration and seeds, now saving the early-stopped best weights and the per-fold standardisation. This covers all 5 folds × 3 seeds for set (a), DFN and CNN1D, for set (d), CNN1D only, and for the shape-only set (e), DFN and CNN1D. The shape-only models were trained with the final protocol and weights were saved in the same run (`SAVEW=1`), so (e) uses the very models of the benchmark.

- The retrained DFN (a), CNN1D (d), and the shape-only runs reproduced the benchmark exactly (test R^2 0.644 ± 0.011 and 0.739 ± 0.010).
- CNN1D (a) differed slightly (0.737 ± 0.009 vs. 0.736 ± 0.012) because of non-deterministic CPU threading in the original runs.

Relevance was computed on the 1,335 held-out test rows for each of the 15 models of a column and averaged over models per row.

Aggregation.

1. **Feature type:** one-hot; each `avg` parameter (SHIFT, SLIDE, RISE, TILT, ROLL, TWIST, the column names of `avg_tetramer.dat`); `fce` diagonal (6 terms); and `fce` off-diagonal (15 terms), each summed over windows.
2. **Position:** one-hot relevance per base position (1–10) and shape relevance per tetramer window (1–7). For CNN1D, the one-hot channel of base k in window i is credited to base position $i{+}k$.
3. **Base×position maps:** the relevance of each nucleotide channel at each position, averaged over the top-50 test k-mers by affinity and over all test rows. We also draw sequence logos for the top-6 test k-mers, with letter height equal to the summed relevance of the 4 channels at that position. All 4 channels are included because z-scored absent-base channels are non-zero.
4. **Shape-only models (e): projected base-level map.** Without one-hot channels there is no input feature that belongs to a single base, so base-level maps are *derived*: the summed relevance of each tetramer window i (all 27 shape channels) is divided equally over its 4 bases $i{+}1..i{+}4$ and placed on the channel of the base that is actually present in that window of the k-mer. Position and heat/logo plots of (e) are therefore a spread-out version of the window relevance. They allow comparison with (a) and (d) but carry no information beyond window relevance, and the ends of the 10-mer (covered by one window) get little by construction.
5. **Signed vs. absolute:** absolute shares $\overline{|R|}_g / \sum_g \overline{|R|}_g$ measure how much a group is used. Signed means over all rows and over the top-100 k-mers show the direction of its effect.

3 Results

3.1 Characterisation of the new shape tables

The tables are complete (all k-mers, no missing values). The `avg` files are tab-separated with a `SEQUENCE` header, and the `fce` files are space-separated. Reverse-complement symmetry holds:

correlation ≥ 0.998 between a k-mer and its reverse complement, with shift and tilt sign-flipped. The new tetramer values differ from the earlier `data/input` tables. Correlations per average parameter are 0.83 (shift), 0.97, 0.96, 0.85 (tilt), 0.98 and 0.94, and 0.90–0.97 for the stiffness diagonal. Hexamer averages correlate 0.96–0.99 with those of their central tetramer, so they are largely redundant. In `4.txt`, log affinities of the forward and reverse-complement orientations of the same 10-mer correlate only $r = 0.417$ (17,376 pairs). The correlation stays at 0.412 and 0.425 when both counts are $\geq$10 or $\geq$30, so the asymmetry is not explained by counting noise.

3.2 Preprocessing ablation

Table 2: Preprocessing ablation: 5-fold grouped CV R^2 on log affinity, evaluated on per-strand count$\geq$10 rows (mean±sd over folds where given). Base: log, count$\geq$10, per-strand, tetramer avg+fce, standardised.

Change vs. base	RF	CNN1D
Base	0.588±0.015	0.692±0.018
Raw label	0.579	0.691
Count $\geq$1 / $\geq$3 / $\geq$30	−1.449 / 0.044 / −0.227	– / – / 0.158
Merged labels + RC augmentation	0.438	0.520
Hexamer only / tetramer+hexamer	0.546 / 0.579	0.578 / 0.672
avg only / fce only	0.560 / 0.586	0.661 / 0.680
No standardisation	0.588	0.653
+ one-hot	**0.599±0.014**	0.738±0.022
One-hot only (no shape)	0.563	**0.739±0.018**
+ one-hot, tetramer+hexamer	0.589	0.721

The low and high count thresholds give strongly negative or near-zero R^2. Their Pearson correlations stay at 0.67–0.76, so the loss comes mainly from biased predicted levels: low-count labels shrink predictions, and a threshold of 30 removes low-affinity training examples. Per-strand labels beat merged labels by 0.15–0.17. The chosen configuration is log label, count $\geq$10, per-strand, tetramer avg+fce plus one-hot, standardised.

3.3 Model comparison per feature set

Tables 3–9 give the full metrics of every model for each of the seven feature sets: (a)–(c) with one-hot, (d) one-hot only, (e)–(g) MD shape only. Set (a), tetramer shape + one-hot, is the preprocessing chosen in the ablation. Learning curves of all neural-network runs for every set are in Appendix A.

Table 3: Feature set (a) tetramer shape + one-hot: all models, same labels, folds and held-out test set. R^2/Pearson/Spearman on log affinity, mean±sd over runs (5 folds; ×3 seeds for NNs). Gap = train − test R^2. Avg-runs = test R^2 of the averaged prediction of all runs of the model. Bold: best test R^2 in this set.

Model	Runs	Train R^2	Val R^2	Test R^2	Test Pearson	Test Spearman	Gap	Avg-runs R^2
Linear	5	0.502±0.002	0.480±0.010	0.471±0.001	0.687±0.001	0.692±0.001	0.031±0.003	0.473
Ridge	5	0.500±0.002	0.480±0.010	0.470±0.001	0.686±0.001	0.692±0.001	0.030±0.003	0.472
SVR	5	0.743±0.002	0.624±0.007	0.598±0.006	0.774±0.004	0.767±0.004	0.146±0.005	0.606
RF	5	0.884±0.001	0.599±0.015	0.572±0.004	0.762±0.003	0.744±0.003	0.311±0.003	0.581
DFN	15	0.820±0.026	0.660±0.011	0.644±0.011	0.803±0.007	0.791±0.008	0.176±0.020	0.673
CNN1D	15	0.889±0.032	0.735±0.015	**0.736±0.012**	0.860±0.006	0.844±0.007	0.153±0.022	0.788
CNNAttention	15	0.880±0.029	0.728±0.019	0.725±0.016	0.853±0.009	0.837±0.010	0.155±0.017	0.778
Transformer	15	0.761±0.018	0.705±0.015	0.685±0.009	0.828±0.006	0.813±0.006	0.077±0.010	0.709

Table 4: Feature set (b) hexamer shape + one-hot: all models, same labels, folds and held-out test set. R^2/Pearson/Spearman on log affinity, mean±sd over runs (5 folds; ×3 seeds for NNs). Gap = train − test R^2. Avg-runs = test R^2 of the averaged prediction of all runs of the model. Bold: best test R^2 in this set.

Model	Runs	Train R^2	Val R^2	Test R^2	Test Pearson	Test Spearman	Gap	Avg-runs R^2
Linear	5	0.474±0.002	0.459±0.011	0.430±0.003	0.656±0.002	0.668±0.001	0.045±0.004	0.431
Ridge	5	0.474±0.002	0.459±0.011	0.431±0.002	0.657±0.001	0.669±0.001	0.042±0.004	0.432
SVR	5	0.725±0.002	0.613±0.009	0.586±0.006	0.767±0.004	0.766±0.004	0.138±0.004	0.594
RF	5	0.881±0.002	0.576±0.017	0.549±0.006	0.745±0.004	0.726±0.004	0.332±0.005	0.558
DFN	15	0.813±0.032	0.652±0.015	0.633±0.016	0.796±0.010	0.785±0.010	0.180±0.020	0.663
CNN1D	15	0.876±0.024	0.717±0.021	**0.714±0.009**	0.847±0.005	0.830±0.006	0.162±0.020	0.765
CNNAttention	15	0.866±0.045	0.709±0.018	0.702±0.016	0.840±0.009	0.825±0.009	0.164±0.036	0.758
Transformer	15	0.763±0.013	0.700±0.012	0.673±0.012	0.821±0.007	0.807±0.008	0.089±0.012	0.699

Table 5: Feature set (c) tetramer+hexamer shape + one-hot: all models, same labels, folds and held-out test set. R^2/Pearson/Spearman on log affinity, mean±sd over runs (5 folds; ×3 seeds for NNs). Gap = train − test R^2. Avg-runs = test R^2 of the averaged prediction of all runs of the model. Bold: best test R^2 in this set.

Model	Runs	Train R^2	Val R^2	Test R^2	Test Pearson	Test Spearman	Gap	Avg-runs R^2
Linear	5	0.533±0.003	0.500±0.014	0.484±0.003	0.696±0.002	0.702±0.002	0.049±0.005	0.487
Ridge	5	0.529±0.003	0.501±0.012	0.483±0.002	0.696±0.001	0.701±0.001	0.046±0.004	0.486
SVR	5	0.730±0.002	0.616±0.007	0.588±0.006	0.768±0.004	0.764±0.004	0.142±0.005	0.596
RF	5	0.893±0.001	0.589±0.018	0.563±0.005	0.755±0.003	0.736±0.003	0.330±0.004	0.572
DFN	15	0.819±0.028	0.657±0.013	0.641±0.009	0.801±0.006	0.789±0.008	0.178±0.022	0.670
CNN1D	15	0.880±0.035	0.722±0.019	**0.723±0.013**	0.852±0.008	0.836±0.008	0.157±0.026	0.773
CNNAttention	15	0.864±0.046	0.713±0.012	0.709±0.015	0.844±0.009	0.830±0.009	0.156±0.037	0.763
Transformer	15	0.762±0.017	0.699±0.012	0.681±0.014	0.826±0.008	0.812±0.009	0.081±0.009	0.707

Table 6: Feature set (d) one-hot only (no shape): all models, same labels, folds and held-out test set. R^2/Pearson/Spearman on log affinity, mean±sd over runs (5 folds; ×3 seeds for NNs). Gap = train − test R^2. Avg-runs = test R^2 of the averaged prediction of all runs of the model. Bold: best test R^2 in this set.

Model	Runs	Train R^2	Val R^2	Test R^2	Test Pearson	Test Spearman	Gap	Avg-runs R^2
Linear	5	0.401±0.003	0.398±0.011	0.380±0.001	0.616±0.001	0.634±0.001	0.021±0.003	0.380
Ridge	5	0.401±0.003	0.398±0.011	0.380±0.001	0.616±0.000	0.634±0.001	0.021±0.003	0.380
SVR	5	0.732±0.002	0.603±0.005	0.581±0.005	0.763±0.003	0.759±0.003	0.150±0.003	0.590
RF	5	0.745±0.003	0.564±0.011	0.536±0.005	0.746±0.002	0.737±0.002	0.209±0.004	0.541
DFN	15	0.810±0.022	0.656±0.012	0.635±0.012	0.798±0.007	0.785±0.007	0.176±0.021	0.663
CNN1D	15	0.897±0.023	0.742±0.022	**0.739±0.010**	0.861±0.005	0.843±0.006	0.159±0.020	0.789
CNNAttention	15	0.909±0.025	0.733±0.021	0.729±0.009	0.856±0.005	0.838±0.005	0.180±0.023	0.785
Transformer	15	0.757±0.017	0.703±0.013	0.679±0.012	0.825±0.007	0.809±0.009	0.078±0.007	0.702

Table 7: Feature set (e) tetramer MD shape only: all models, same labels, folds and held-out test set. R^2/Pearson/Spearman on log affinity, mean±sd over runs (5 folds; ×3 seeds for NNs). Gap = train − test R^2. Avg-runs = test R^2 of the averaged prediction of all runs of the model. Bold: best test R^2 in this set.

Model	Runs	Train R^2	Val R^2	Test R^2	Test Pearson	Test Spearman	Gap	Avg-runs R^2
Linear	5	0.452±0.003	0.431±0.014	0.416±0.001	0.645±0.001	0.649±0.002	0.036±0.004	0.418
Ridge	5	0.451±0.003	0.431±0.013	0.415±0.001	0.645±0.001	0.648±0.002	0.035±0.004	0.417
SVR	5	0.691±0.002	0.576±0.006	0.547±0.007	0.741±0.005	0.734±0.006	0.144±0.006	0.555
RF	5	0.880±0.001	0.588±0.016	0.562±0.005	0.754±0.003	0.734±0.003	0.317±0.004	0.571
DFN	15	0.807±0.026	0.621±0.012	0.606±0.011	0.779±0.007	0.764±0.007	0.201±0.021	0.638
CNN1D	15	0.866±0.028	0.694±0.021	**0.700±0.009**	0.838±0.006	0.821±0.006	0.167±0.024	0.751
CNNAttention	15	0.858±0.038	0.683±0.020	0.683±0.013	0.829±0.008	0.811±0.009	0.175±0.031	0.742
Transformer	15	0.727±0.014	0.665±0.016	0.649±0.010	0.806±0.006	0.790±0.007	0.077±0.006	0.675

Table 8: Feature set (f) hexamer MD shape only: all models, same labels, folds and held-out test set. R^2/Pearson/Spearman on log affinity, mean±sd over runs (5 folds; ×3 seeds for NNs). Gap = train − test R^2. Avg-runs = test R^2 of the averaged prediction of all runs of the model. Bold: best test R^2 in this set.

Model	Runs	Train R^2	Val R^2	Test R^2	Test Pearson	Test Spearman	Gap	Avg-runs R^2
Linear	5	0.339±0.004	0.323±0.019	0.314±0.002	0.561±0.002	0.557±0.002	0.025±0.004	0.316
Ridge	5	0.339±0.004	0.323±0.019	0.314±0.002	0.561±0.002	0.557±0.002	0.025±0.004	0.315
SVR	5	0.581±0.003	0.480±0.013	0.462±0.004	0.681±0.003	0.677±0.004	0.119±0.003	0.469
RF	5	0.871±0.002	0.547±0.020	0.521±0.007	0.724±0.005	0.697±0.005	0.350±0.005	0.532
DFN	15	0.749±0.034	0.540±0.021	0.529±0.010	0.728±0.007	0.710±0.008	0.220±0.030	0.565
CNN1D	15	0.756±0.049	0.572±0.029	0.553±0.011	0.747±0.008	0.728±0.008	0.203±0.042	0.619
CNNAttention	15	0.770±0.034	0.573±0.025	0.553±0.010	0.748±0.007	0.732±0.007	0.218±0.035	0.613
Transformer	15	0.652±0.021	0.584±0.021	**0.563±0.014**	0.751±0.010	0.732±0.012	0.089±0.010	0.590

Table 9: Feature set (g) tetramer+hexamer MD shape only: all models, same labels, folds and held-out test set. R^2/Pearson/Spearman on log affinity, mean±sd over runs (5 folds; ×3 seeds for NNs). Gap = train − test R^2. Avg-runs = test R^2 of the averaged prediction of all runs of the model. Bold: best test R^2 in this set.

Model	Runs	Train R^2	Val R^2	Test R^2	Test Pearson	Test Spearman	Gap	Avg-runs R^2
Linear	5	0.493±0.003	0.461±0.015	0.443±0.002	0.667±0.001	0.669±0.001	0.051±0.004	0.446
Ridge	5	0.489±0.003	0.461±0.013	0.441±0.002	0.665±0.001	0.667±0.001	0.048±0.004	0.444
SVR	5	0.678±0.002	0.565±0.008	0.536±0.006	0.733±0.004	0.729±0.005	0.142±0.005	0.544
RF	5	0.890±0.001	0.579±0.018	0.553±0.005	0.747±0.004	0.726±0.004	0.337±0.004	0.563
DFN	15	0.809±0.035	0.616±0.014	0.601±0.012	0.776±0.008	0.761±0.009	0.209±0.031	0.633
CNN1D	15	0.844±0.044	0.670±0.024	**0.672±0.016**	0.822±0.009	0.806±0.010	0.172±0.034	0.729
CNNAttention	15	0.837±0.047	0.667±0.024	0.664±0.019	0.817±0.012	0.802±0.013	0.173±0.035	0.725
Transformer	15	0.734±0.013	0.664±0.016	0.644±0.008	0.804±0.005	0.790±0.006	0.090±0.008	0.671

Table 10: Training dynamics of the neural networks with feature set (a) (means over runs; MSE on standardised log affinity).

Model	Best epoch	Epochs run	Final train MSE	Best val MSE
DFN	44.3	70.3	0.186	0.331
CNN1D	27.1	53.1	0.068	0.261
CNNAttention	21.6	47.6	0.055	0.266
Transformer	111.6	137.6	0.293	0.291

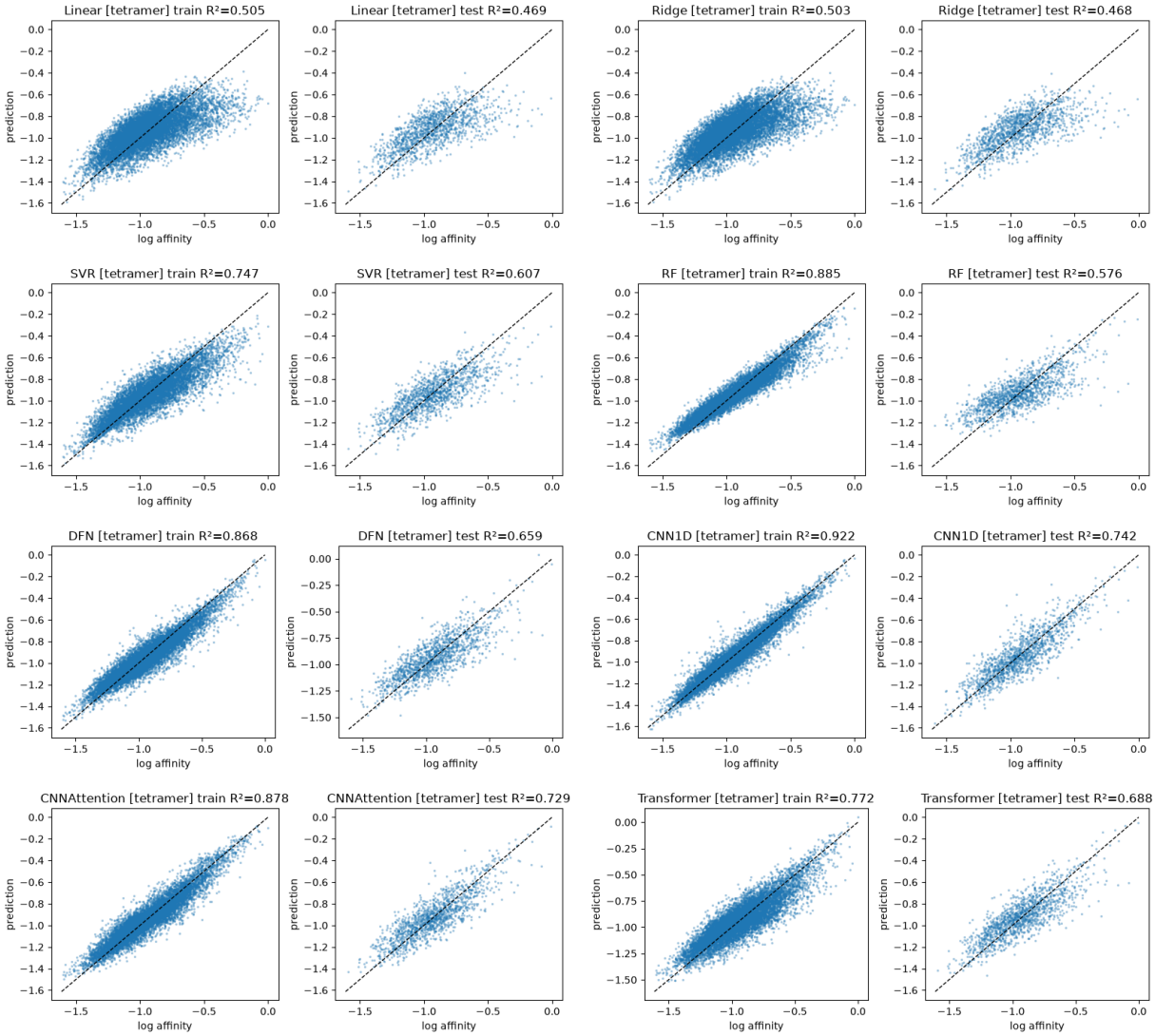


Figure 2: Ground truth vs. prediction (fold 0, seed 0) for each model with feature set (a) on its training fold (left panel) and on the held-out test set (right panel), with R^2.

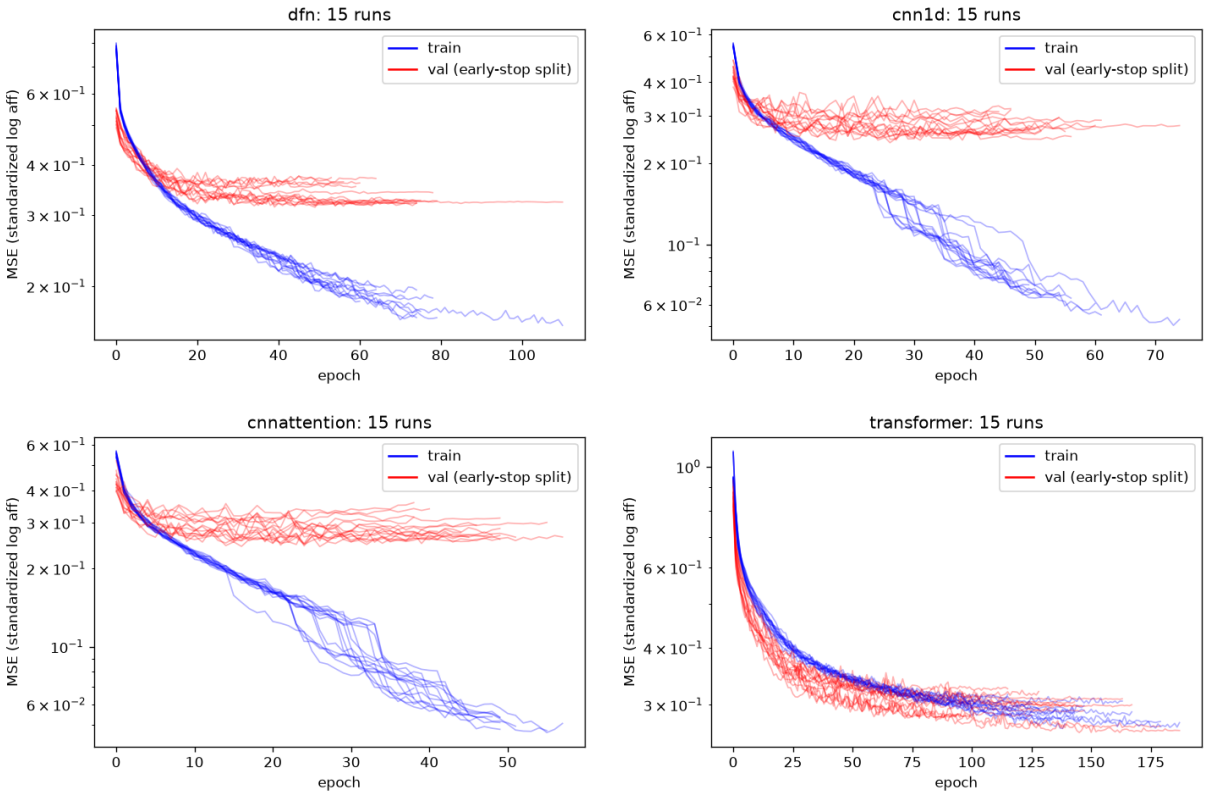


Figure 3: Learning curves of all neural-network runs with feature set (a) (blue: training loss, red: early-stopping validation loss).

3.4 Summary: with vs. without one-hot

Tables 3–9 give every metric. Table 11 puts the test R^2 of the shape sets side by side with and without one-hot (a vs. e, b vs. f, c vs. g), plus one-hot only (d).

Table 11: Test R^2 (mean over runs) per model and feature set. Columns: shape + one-hot (a, b, c) next to the same shape only (e, f, g), then one-hot only (d). Bold: best set per model.

Model	tetramer (a) +1H	tetramer (e) MD only	hexamer (b) +1H	hexamer (f) MD only	tetra+hexa (c) +1H	tetra+hexa (g) MD only	one-hot (d)
Linear	0.471	0.416	0.430	0.314	**0.484**	0.443	0.380
Ridge	0.470	0.415	0.431	0.314	**0.483**	0.441	0.380
SVR	**0.598**	0.547	0.586	0.462	0.588	0.536	0.581
RF	**0.572**	0.562	0.549	0.521	0.563	0.553	0.536
DFN	**0.644**	0.606	0.633	0.529	0.641	0.601	0.635
CNN1D	0.736	0.700	0.714	0.553	0.723	0.672	**0.739**
CNNAttention	0.725	0.683	0.702	0.553	0.709	0.664	**0.729**
Transformer	**0.685**	0.649	0.673	0.563	0.681	0.644	0.679

Table 12: Averaged-runs test R^2 (mean prediction of all runs of a model) for the same sets and models as Table 11.

Model	tetramer		hexamer		tetra+hexa		one-hot
	(a) +1H	(e) MD only	(b) +1H	(f) MD only	(c) +1H	(g) MD only	(d)
Linear	0.473	0.418	0.431	0.316	**0.487**	0.446	0.380
Ridge	0.472	0.417	0.432	0.315	**0.486**	0.444	0.380
SVR	**0.606**	0.555	0.594	0.469	0.596	0.544	0.590
RF	**0.581**	0.571	0.558	0.532	0.572	0.563	0.541
DFN	**0.673**	0.638	0.663	0.565	0.670	0.633	0.663
CNN1D	0.788	0.751	0.765	0.619	0.773	0.729	**0.789**
CNNAttention	0.778	0.742	0.758	0.613	0.763	0.725	**0.785**
Transformer	**0.709**	0.675	0.699	0.590	0.707	0.671	0.702

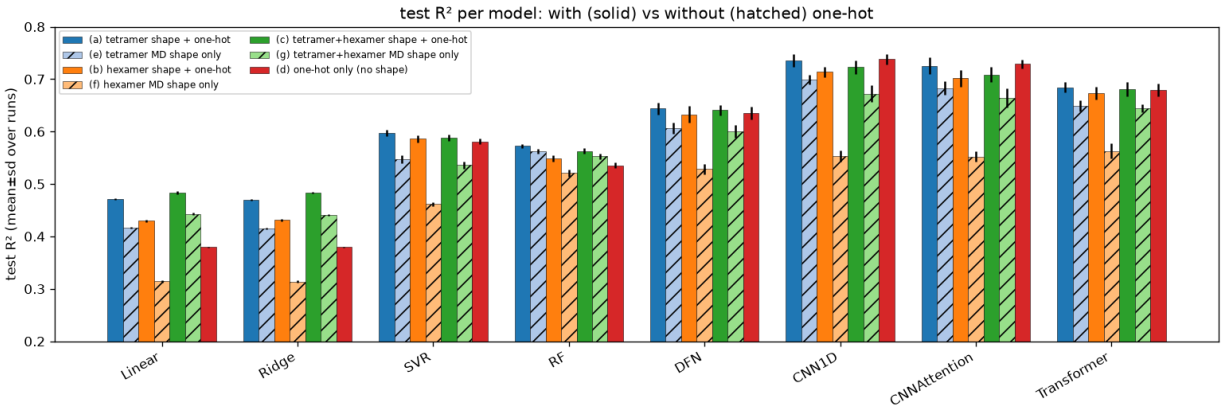


Figure 4: Test R^2 (mean±sd over runs) per model and feature set. Solid bars: shape + one-hot and one-hot only; hatched bars: the same shape without one-hot.

3.5 Can MD-derived features alone predict affinity?

This is the central question of the study. Sets (e)–(g) use only the MD tables, with no explicit sequence input.

Table 13: Tetramer MD shape only (set e) compared with one-hot only (d) and MD + one-hot (a). Test R^2 (mean over runs); (e) with sd, Pearson, Spearman and averaged-runs R^2. Δ = (e) − (d); ratio = (e)/(d).

Model	(d) one-hot	(a) MD+1H	(e) MD only	Δ(e−d)	ratio	Pearson (e)	Spearman (e)	Avg-runs (e)
Linear	0.380	0.471	0.416±0.001	+0.036	1.10	0.645	0.649	0.418
Ridge	0.380	0.470	0.415±0.001	+0.035	1.09	0.645	0.648	0.417
SVR	0.581	0.598	0.547±0.007	-0.034	0.94	0.741	0.734	0.555
RF	0.536	0.572	0.562±0.005	+0.027	1.05	0.754	0.734	0.571
DFN	0.635	0.644	0.606±0.011	-0.029	0.95	0.779	0.764	0.638
CNN1D	0.739	0.736	0.700±0.009	-0.039	0.95	0.838	0.821	0.751
CNNAttention	0.729	0.725	0.683±0.013	-0.046	0.94	0.829	0.811	0.742
Transformer	0.679	0.685	0.649±0.010	-0.030	0.96	0.806	0.790	0.675

MD-only vs. one-hot-only vs. MD + one-hot.

- **MD alone predicts affinity well.** The best MD-only model is CNN1D with test $R^2 = 0.700 \pm 0.009$ (Pearson 0.838, Spearman 0.821; 0.751 when its 15 runs are averaged). This is 95% of the one-hot-only value (0.739) and of the MD + one-hot value (0.736). CNNAttention

reaches 0.683, the Transformer 0.649, DFN 0.606, RF 0.562, SVR 0.547 and Linear/Ridge 0.416/0.415.

- **MD-only is not better than sequence for deep models.** For SVR, DFN, CNN1D, CNNAttention and the Transformer, (e) is 0.03–0.05 R^2 below (d). The converse holds for the two models that cannot learn their own sequence code: Linear/Ridge gain +0.036 and RF +0.027 from MD over one-hot. For those models the MD numbers act as useful engineered, non-linear features of the sequence.

- **MD + one-hot adds nothing for the CNNs.** (a) vs. (d): 0.736 vs. 0.739 for CNN1D, 0.725 vs. 0.729 for CNNAttention, so combining does not beat sequence alone for the best models.

- **Which MD set.** Tetramer MD (e) is the best MD-only set for every non-linear model. Hexamer MD only (f) is the worst (CNN1D 0.553, DFN 0.529, Linear 0.314). Tetramer+hexamer MD (g) is below (e) for the NNs (CNN1D 0.672) and above it only for Linear/Ridge (0.443/0.441 vs. 0.416/0.415). The hexamer tables provide only 5 windows, so the two outermost tetramer windows of the 10-mer have no hexamer information.

How much sequence identity is in the MD features? The 256 tetramer vectors have full rank 27 and are well separated (smallest pairwise distance 0.75 in z-units, median 7.4), so a model can in principle use them as labels of the tetramer. Nearest-neighbour recovery of all 256 tetramers stays at 100% after adding noise of sd 0.1. A linear classifier on the 27 values predicts the two *central* bases of a tetramer with 100% cross-validated accuracy, and the two *outer* bases with 75.8% (chance 25%). The shape tables therefore encode nearly all sequence information, and the models are free to use them as identity codes.

Table 14: Permuted-table control on set (e). Test R^2 (mean±sd over runs) with the real MD table and with two random reassignments of the MD rows to tetramers (perm. 1, 2); difference of the real table to the mean of the two permutations; train − test gap; averaged-runs R^2 and its difference; one-hot only (d) for reference.

	test R^2			Δ	gap		avg-runs R^2			(d)
Model	real MD	perm. 1	perm. 2	real − perm.	real	perm. (mean)	real	perm. (mean)	Δ	test R^2
RF	0.562±0.005	0.505±0.006	0.516±0.005	+0.052	0.317	0.369	0.571	0.519	+0.052	0.536
DFN	0.606±0.011	0.472±0.014	0.470±0.010	+0.135	0.201	0.361	0.638	0.526	+0.112	0.635
CNN1D	0.700±0.009	0.675±0.010	0.681±0.010	+0.022	0.167	0.227	0.751	0.747	+0.005	0.739

Control: real MD values vs. a random lookup table. The permuted table has the same dimension, value distribution and identifying power as the real one, but the values are unrelated to the tetramer's physics. Result:

- **DFN:** real 0.606 vs. 0.472/0.470 for the permuted tables (+0.135). The real MD values are worth far more than an arbitrary code, plausibly because physically similar tetramers get similar inputs, which a feedforward network can exploit as smooth features (we did not test this explanation directly).

- **RF:** real 0.562 vs. 0.505/0.516 (+0.052).

- **CNN1D:** real 0.700 vs. 0.675/0.681 (+0.022, about two sd over runs). With a random table, the CNN still reaches 0.678, because its first layer can learn any tetramer-wise lookup, as it does from one-hot. The averaged-runs difference is only +0.005 (0.751 vs. 0.744/0.750).

- The permuted tables also overfit more (gap 0.23–0.37 vs. 0.17–0.32 for the real table, with the largest difference for DFN: 0.361 vs. 0.201).

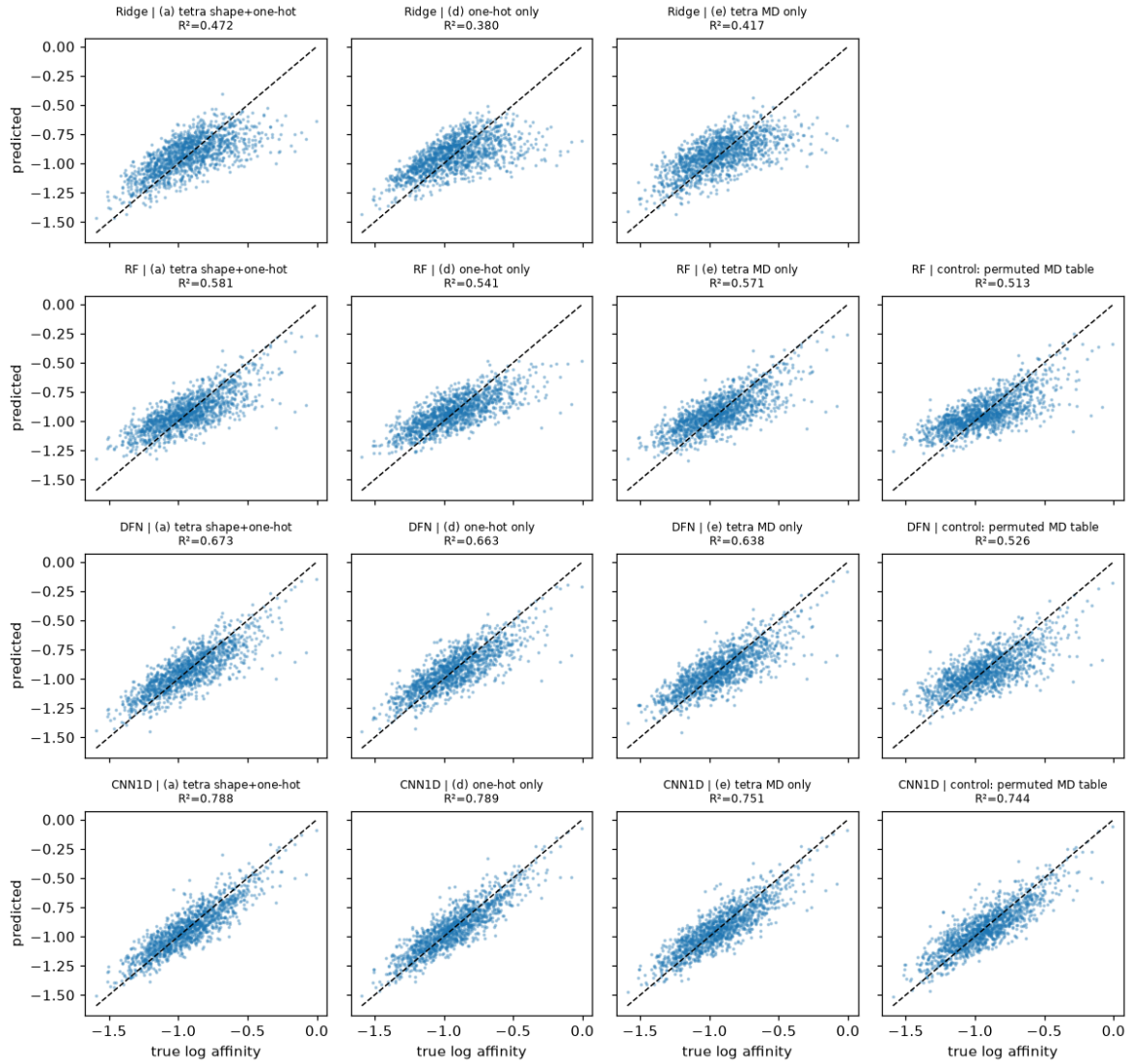


Figure 5: True vs. predicted log affinity on the held-out test set (mean prediction of all runs of a model, so the R^2 is the averaged-runs value) for Ridge, RF, DFN and CNN1D: tetramer MD + one-hot (a), one-hot only (d), tetramer MD only (e) and the permuted-table control (RF, DFN, CNN1D).

Verdict. Good affinity prediction is possible from MD-derived data alone: with tetramer shape and stiffness only, CNN1D reaches test $R^2 = 0.700$ (Pearson 0.84), at most 0.04–0.05 below one-hot sequence encoding and MD + one-hot. The MD tables, however, are a complete tetramer lookup that identifies the sequence (central bases 100%, outer bases 76% linearly decodable), and they never beat the sequence encoding for the deep models. The permuted-table control shows that the physical content of the values is not irrelevant, since it is worth +0.135 for DFN, +0.052 for RF and +0.022 for CNN1D, but for the best model (CNN1D) about 97% of the MD-only R^2 (0.678 of 0.700) is reachable with an arbitrary injective code of the tetramer. So the data show that MD-derived features *contain* enough information to predict Gata4 affinities, and that the sequence information they carry is the main source of this. They do not show that the networks rely on physical deformability beyond what a sequence code provides. For models that cannot learn their own code (Linear, RF, DFN), the real MD values are a clearly better feature set than one-hot or random codes.

3.6 Processing time

Table 15: Processing time for sets (a)–(d): feature building for all 13,100 rows [s], training time per run [s] (mean±sd), and inference [s per 1,000 sequences; 0.00 means <0.005 s]. Tetramer (a) timings come from the re-timing runs (seed 0, 5 folds); the other sets were timed on all their runs. Hardware and threading are described in the timing subsection of the Methods.

	(a) tetramer + 1H			(b) hexamer + 1H			(c) tetra+hexa + 1H			(d) one-hot only		
Model	feat	train/run	inf/1k	feat	train/run	inf/1k	feat	train/run	inf/1k	feat	train/run	inf/1k
Linear	0.42	8.4±0.9	0.00	0.27	6.0±0.5	0.00	0.55	13.1±1.3	0.02	0.03	0.4±0.1	0.00
Ridge	0.57	0.6±0.0	0.00	0.12	0.3±0.3	0.00	0.46	0.9±0.2	0.02	0.03	0.0±0.0	0.00
SVR	0.39	58.6±7.3	2.98	0.50	62.1±10.0	3.19	0.20	84.5±19.0	4.00	0.03	10.6±0.8	0.48
RF	0.44	42.6±2.5	0.21	0.12	46.6±3.6	0.29	0.91	80.6±12.5	0.18	0.03	2.3±0.3	0.10
DFN	0.11	17.6±4.3	0.00	0.10	17.1±3.4	0.00	0.17	17.4±3.3	0.00	0.03	15.8±2.4	0.00
CNN1D	0.36	50.0±8.1	0.01	0.47	49.0±5.9	0.01	0.47	46.5±5.9	0.01	0.19	43.6±5.7	0.01
CNNAttention	0.35	55.4±8.2	0.06	0.74	76.2±16.3	0.06	0.49	55.7±10.9	0.04	0.23	58.2±6.1	0.04
Transformer	0.34	134.6±22.4	0.02	0.85	329.5±60.6	0.04	0.51	131.0±29.9	0.02	0.42	136.1±30.3	0.02

Table 16: Processing time for the shape-only sets (e)–(g), same columns and conventions as Table 15.

	(e) tetramer MD only			(f) hexamer MD only			(g) tetra+hexa MD only		
Model	feat	train/run	inf/1k	feat	train/run	inf/1k	feat	train/run	inf/1k
Linear	0.15	3.9±0.3	0.00	0.14	2.6±0.1	0.00	0.35	5.4±0.5	0.00
Ridge	0.26	0.2±0.1	0.00	0.10	0.2±0.1	0.00	0.39	0.5±0.1	0.00
SVR	0.09	30.5±5.3	1.56	0.07	24.0±4.2	1.32	0.19	27.8±8.4	1.20
RF	0.08	24.2±2.1	0.13	0.07	23.6±1.7	0.12	0.15	30.1±8.4	0.08
DFN	0.09	25.1±5.9	0.00	0.07	29.9±6.3	0.00	0.17	23.4±6.0	0.00
CNN1D	0.15	52.3±7.7	0.01	0.26	52.0±12.5	0.01	0.28	53.2±9.0	0.01
CNNAttention	0.15	62.8±8.8	0.05	0.22	61.9±7.4	0.05	0.32	60.4±7.7	0.05
Transformer	0.15	150.6±27.2	0.02	0.30	166.6±30.2	0.02	0.51	164.4±32.6	0.02

Table 17: Total training time [min] for the full protocol (5 runs for non-NN models, 15 for NNs) = mean time per run × number of runs.

Model	(a)	(b)	(c)	(d)	(e)	(f)	(g)
Linear	0.7	0.5	1.1	0.0	0.3	0.2	0.4
Ridge	0.1	0.0	0.1	0.0	0.0	0.0	0.0
SVR	4.9	5.2	7.0	0.9	2.5	2.0	2.3
RF	3.6	3.9	6.7	0.2	2.0	2.0	2.5
DFN	4.4	4.3	4.4	4.0	6.3	7.5	5.8
CNN1D	12.5	12.2	11.6	10.9	13.1	13.0	13.3
CNNAttention	13.9	19.1	13.9	14.5	15.7	15.5	15.1
Transformer	33.7	82.4	32.7	34.0	37.7	41.6	41.1

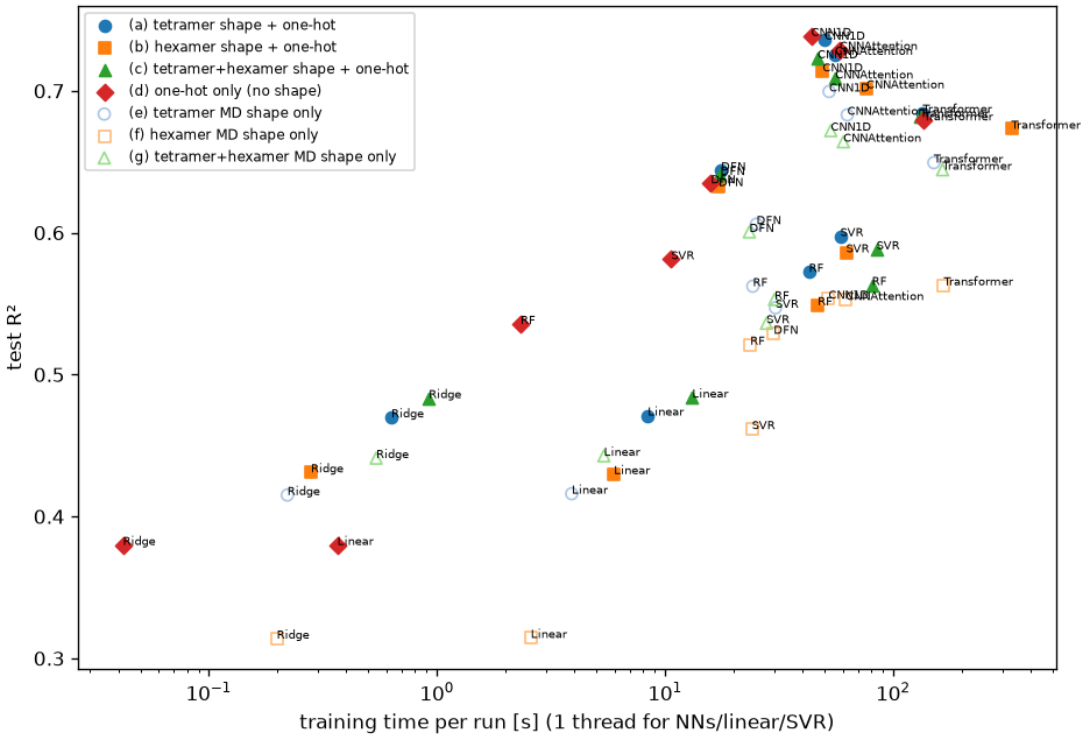


Figure 6: Training time per run vs. test R^2 for every model and feature set (filled: with one-hot; open markers: shape only).

3.7 Which MD parameters drive the predictions? Feature importance (LRP)

LRP was computed for DFN and CNN1D on set (a) (shape + one-hot), CNN1D on (d) (one-hot only) and, for the MD-aim, DFN and CNN1D on (e) (tetramer MD only). Everything below uses 15 models per column (5 folds × 3 seeds), the same rules and aggregations. Table 18 shows the sanity checks.

Table 18: LRP sanity checks on the test set (15 models each). Corr/slope: Pearson correlation and slope of $\sum_j R_j$ vs. $f(x)$. Ratio: median $\sum_j R_j / f(x)$ for $|f(x)| > 0.5$. LRP-0 vs. G×I: maximum relative deviation $\max|R^{(0)} - G{\times}I| / \max|G{\times}I|$ over runs. Corr G×I: correlation of the composite LRP maps with gradient×input.

Model (set)	Runs	Corr	Slope	Ratio	LRP-0 vs. G×I	Corr G×I
DFN (a)	15	0.987	0.759	0.718	1.3e-04	0.984
CNN1D (a)	15	0.550	0.510	0.473	3.5e-02	0.166
CNN1D (d)	15	0.557	0.532	0.502	3.5e-04	0.144
DFN (e)	15	0.980	0.710	0.653	2.6e-05	0.983
CNN1D (e)	15	0.568	0.521	0.514	1.1e-02	0.169

LRP-0 matched gradient×input to 10^{-4} or better for the DFNs and CNN1D (d), to 1.1% for CNN1D (e), and to 3.5% for CNN1D (a), where some near-zero denominators in the residual split amplify rounding. This confirms the implementation. Conservation is close for the DFN (corr 0.980–0.987, 65–72% of the output recovered by the median ratio). For CNN1D, about half of the output is recovered and the correlation is 0.55–0.57, because a large part of the relevance is absorbed by the folded BatchNorm biases. The composite γ rule also changes the CNN maps substantially relative to gradient×input (corr 0.14–0.17), whereas they are nearly identical for the DFN (0.984).

Table 19: Share of total absolute LRP relevance per feature type (test rows, averaged over 15 models). One-hot in (d) is the only input, hence 1.000; "–" = group not in the model's input.

Feature group	DFN (a)	CNN1D (a)	CNN1D (d)	DFN (e)	CNN1D (e)
	with one-hot / one-hot only			MD only	
one-hot	0.347	0.489	1.000	–	–
`avg` SHIFT	0.013	0.022	–	0.027	0.049
`avg` SLIDE	0.019	0.022	–	0.027	0.037
`avg` RISE	0.019	0.020	–	0.030	0.041
`avg` TILT	0.017	0.021	–	0.039	0.058
`avg` ROLL	0.015	0.017	–	0.028	0.030
`avg` TWIST	0.018	0.021	–	0.030	0.035
`fce` diagonal	0.136	0.107	–	0.177	0.179
`fce` off-diagonal	0.417	0.281	–	0.642	0.571

Table 20: Mean signed LRP relevance per feature type (standardised log-affinity units): over all test rows (left five columns) and over the top-100 test k-mers by affinity (right five columns).

Feature group	all test rows					top-100 k-mers				
	DFN (a)	CNN1D (a)	CNN1D (d)	DFN (e)	CNN1D (e)	DFN (a)	CNN1D (a)	CNN1D (d)	DFN (e)	CNN1D (e)
one-hot	-0.002	+0.072	+0.146	–	–	+0.331	+0.525	+1.063	–	–
avg SHIFT	-0.000	-0.000	–	-0.013	-0.014	-0.007	+0.010	–	-0.042	+0.005
avg SLIDE	+0.001	+0.002	–	+0.002	+0.002	+0.027	+0.017	–	+0.055	+0.033
avg RISE	+0.020	+0.011	–	+0.053	+0.024	+0.121	+0.053	–	+0.216	+0.120
avg TILT	-0.007	+0.003	–	-0.010	-0.002	-0.016	+0.017	–	-0.023	+0.038
avg ROLL	-0.012	-0.000	–	-0.018	-0.003	-0.015	+0.005	–	-0.022	+0.009
avg TWIST	-0.008	+0.003	–	-0.007	+0.002	+0.009	+0.029	–	+0.014	+0.044
fce diagonal	-0.026	+0.007	–	-0.025	-0.006	+0.078	+0.075	–	+0.114	+0.129
fce off-diagonal	+0.029	+0.040	–	+0.062	+0.057	+0.464	+0.238	–	+0.644	+0.519

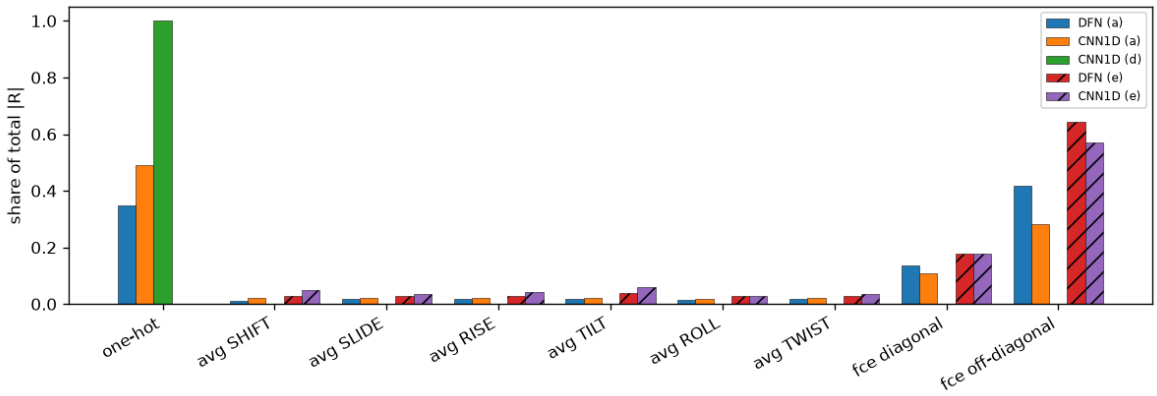


Figure 7: Share of total absolute LRP relevance per feature type for DFN (a), CNN1D (a), CNN1D (d), DFN (e) and CNN1D (e).

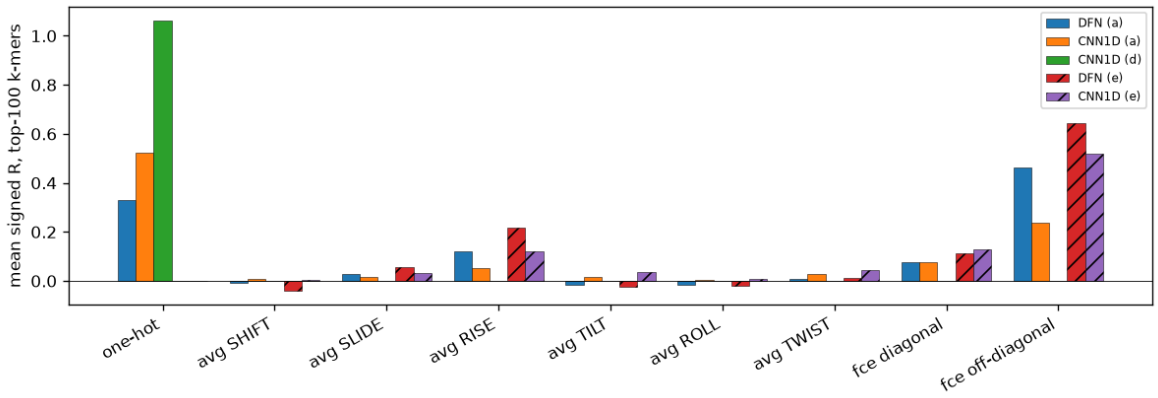


Figure 8: Mean signed LRP relevance per feature type, all test rows and top-100 k-mers, for the same five models.

Feature types: with one-hot.

- **CNN1D (a):** one-hot carries 48.9% of the absolute relevance, `fce` 38.9% (off-diagonal 28.1%, diagonal 10.8%) and `avg` 12.2%.
- **DFN (a):** the DFN relies more on shape: `fce` carries 55.2% (off-diagonal 41.7%), one-hot 34.7% and `avg` 10.1%.

Feature types: MD only (set e).

- **The `fce` (stiffness) block dominates:** `fce` off-diagonal (15 coupling terms per window) carries 64.2% of the relevance in DFN (e) and 57.1% in CNN1D (e), `fce` diagonal 17.7% and 17.9%, and the six `avg` parameters together 18.1% and 25.0%. The off-diagonal couplings are 15 of the 27 channels per window, so per channel they are not more used than the others (0.043 vs. 0.030 per `avg` channel for DFN (e)), yet they are the largest block.
- **`avg` parameters:** none dominates (2.7–3.9% each for DFN (e), 3.0–5.8% for CNN1D (e)). TILT is the largest for both (3.9% and 5.8%), followed by SHIFT in CNN1D (e) (4.9%) and by RISE and TWIST in DFN (e) (3.0% each).
- **Largest `fce` terms:** TILT–ROLL (10.6%), TILT–TWIST (5.5%) and SHIFT–RISE (5.3%) for DFN (e); SLIDE–TILT (6.1%), ROLL–TWIST (5.3%) and TILT–ROLL (5.0%) for CNN1D (e). For (a) they were TILT–ROLL (5.9%), SHIFT–SHIFT (3.6%), RISE–RISE (3.3%) for DFN and SLIDE–TILT (2.8%), ROLL–TWIST (2.2%), SLIDE–ROLL (2.2%) for CNN1D. TILT–ROLL, SLIDE–TILT and ROLL–TWIST (couplings between tilt, roll, twist and slide) recur among the top terms across the models.
- **Direction:** on the top-100 k-mers RISE has the largest positive signed relevance of the `avg` parameters (+0.216 DFN (e), +0.120 CNN1D (e); +0.121 and +0.053 in (a)), and `fce` off-diagonal is strongly positive (+0.644 and +0.519). Over all rows the signed relevances are near zero.
- **Comparison with (a):** without the one-hot channels the shape block must carry the whole sequence signal, so the relevance moves onto `fce` off-diagonal (41.7% → 64.2% for DFN, 28.1% → 57.1% for CNN1D) and `avg` (10.1% → 18.1%, 12.2% → 25.0%).

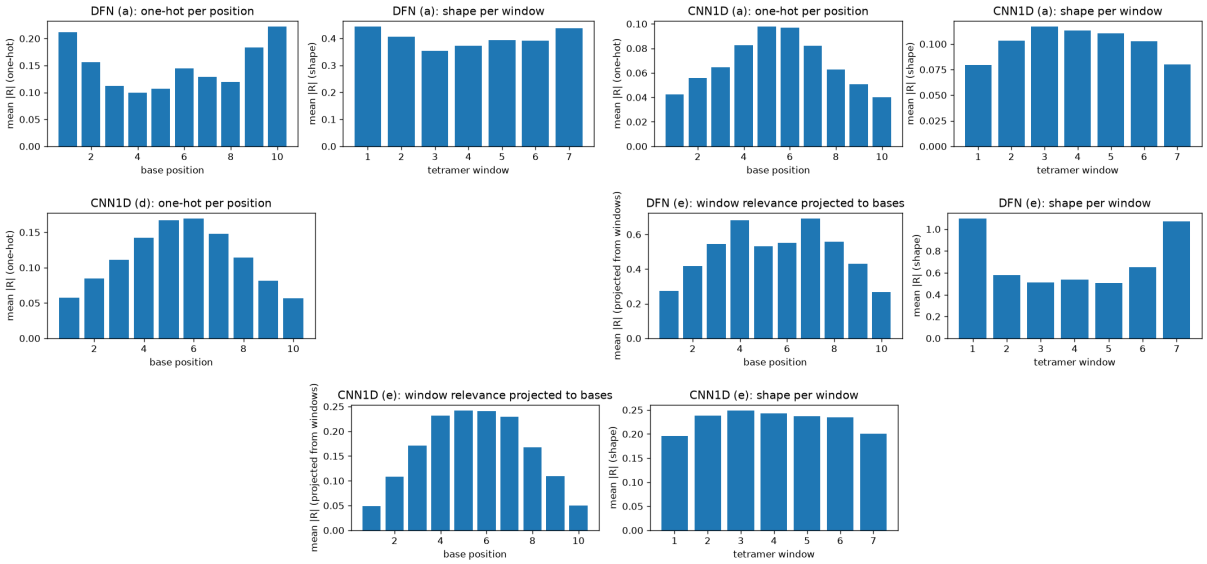


Figure 9: Mean $|R|$ per base position (for sets (a), (d): one-hot relevance; for set (e): projected relevance, see Methods) and per tetramer window (shape relevance).

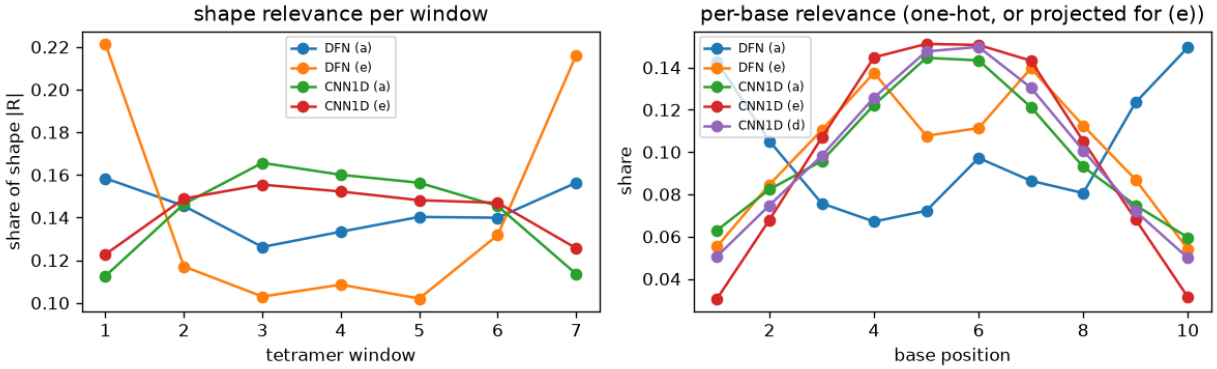


Figure 10: Left: share of shape relevance per tetramer window. Right: share of per-base relevance per base position (one-hot relevance for (a), (d); projected window relevance for (e)).

Positions and windows.

- **CNN1D:** relevance peaks at the centre for one-hot (positions 5–6: 0.098/0.097 vs. 0.043/0.040 at the ends for set (a); 0.167/0.169 vs. 0.057/0.056 for (d)) and for shape (windows 3–5: 0.110–0.117 vs. 0.080 at windows 1 and 7). The shape window profile of CNN1D (e) is flat and slightly centre-peaked (window shares 12.3% at window 1, 15.6% at window 3, 12.6% at window 7). This follows from zero padding: edge positions enter fewer receptive fields.

- **DFN:** the profile is U-shaped. In set (a), one-hot relevance is highest at the end bases (0.21–0.22 vs. 0.10–0.11 at positions 4–5), and shape relevance is highest in windows 1 and 7 (0.44 vs. 0.35–0.39). Without one-hot, DFN (e) depends even more on the end windows: they carry 22.1% and 21.6% of the shape relevance, vs. 10.2–13.2% for the central windows 2–6 (window 6 = 13.2%). The end bases are covered by only one tetramer window, so their identity is only available there, and without one-hot the DFN reads them from the end-window shape.

- **Projection for (e).** The per-base curves for (e) (right panel of Figure 10) are obtained from window relevance by the equal split described in Methods. Base positions 1 and 10

receive only a quarter of one window's relevance by construction, and the central positions several windows, so these curves show the window relevance spread over bases and are not an independent base-level measurement.

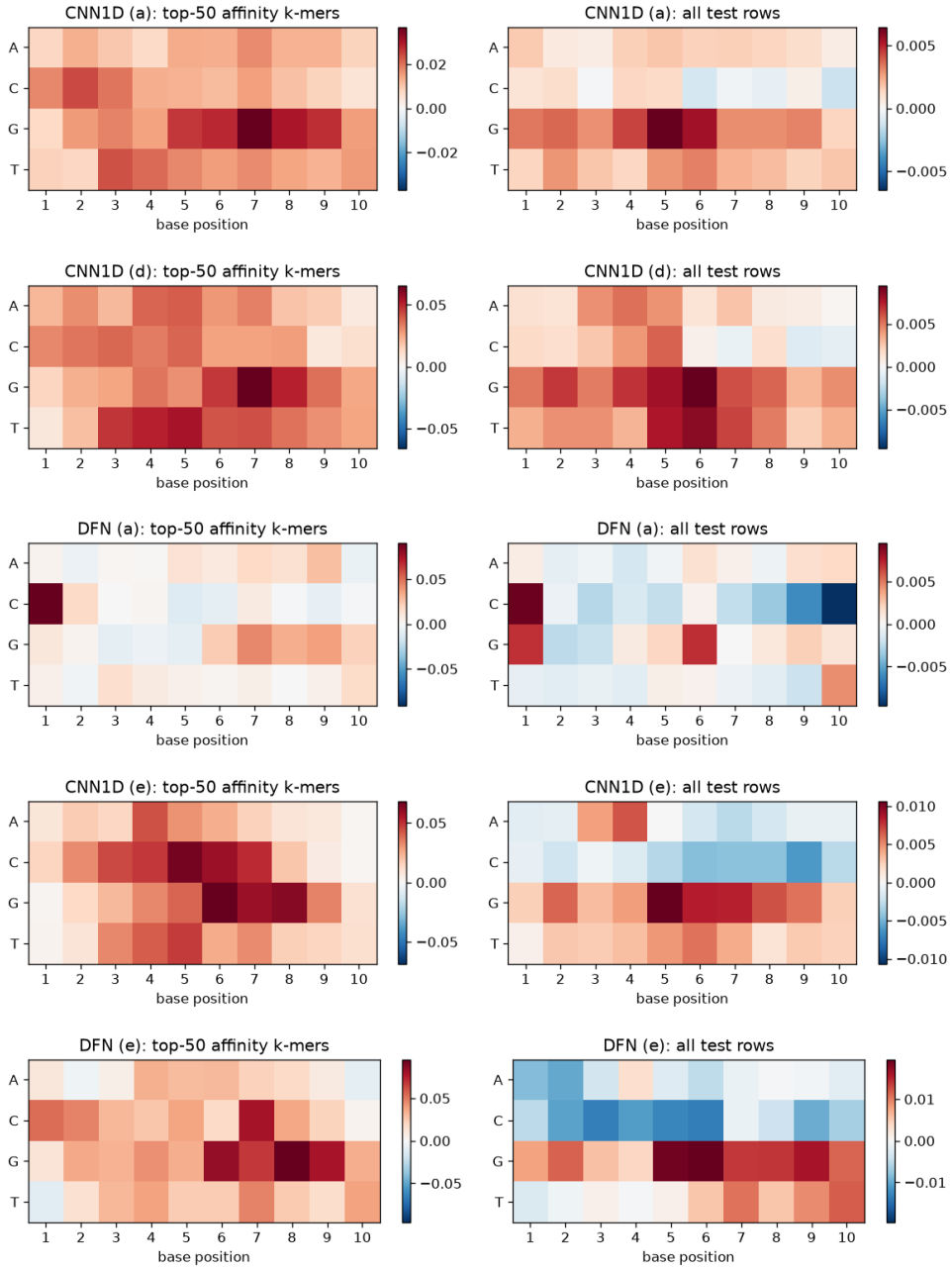


Figure 11: Mean signed LRP relevance per nucleotide channel and base position: top-50 test k-mers by affinity (left) and all test rows (right). For (e), the base-level map is projected from window relevance onto the channel of the base present in each window.

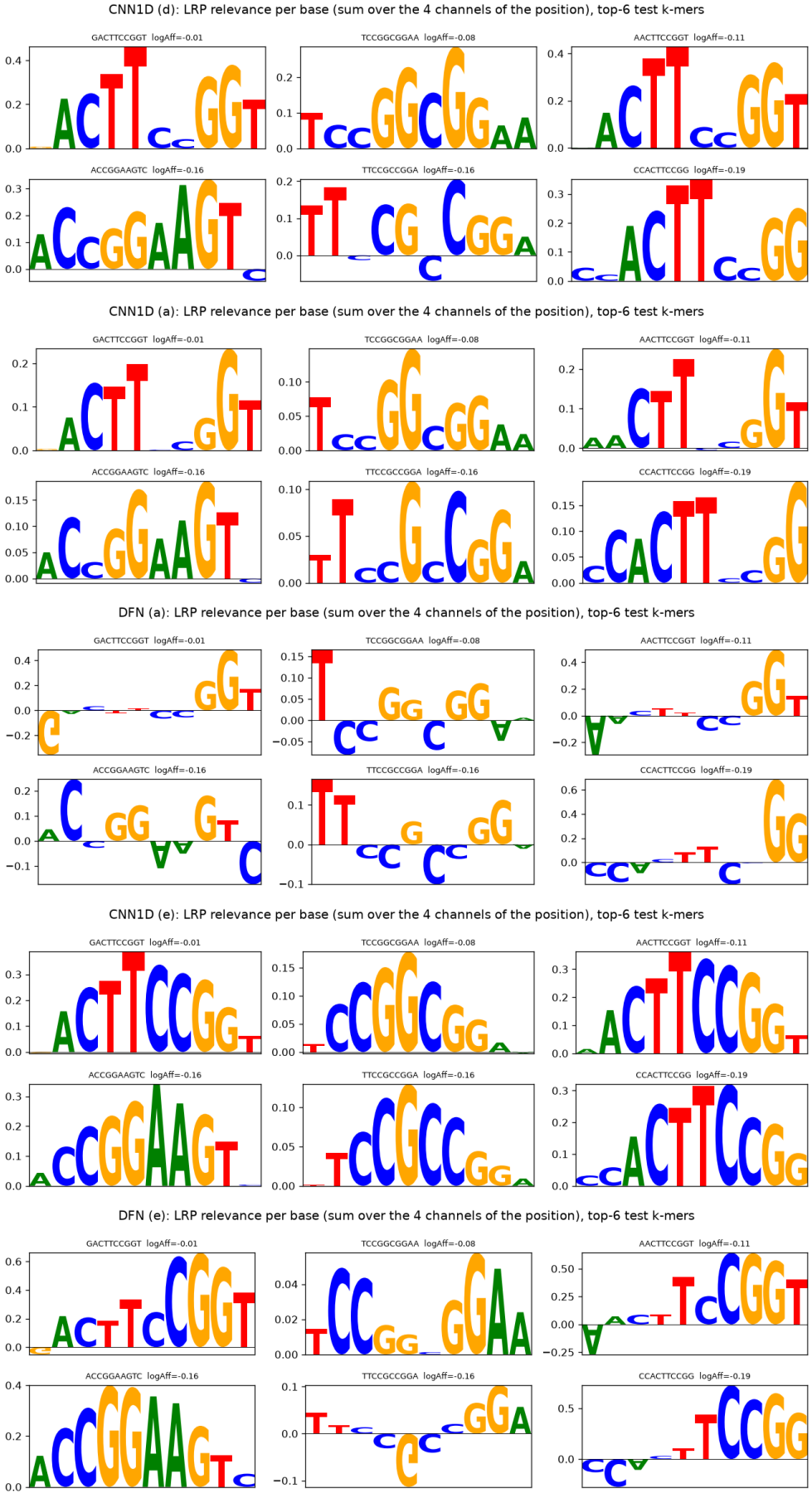


Figure 12: Relevance logos for the six highest-affinity test k-mers (letter height = signed relevance of the position; for (e), projected from window relevance).

Motif check. The highest-affinity k-mers do not contain a GATA site; they are ETS-like (ACTTCCGG / CCGGAAGT). Across all of `4.txt`:

- the 96 k-mers containing CCGGAAGT or ACTTCCGG have mean log affinity -0.29, vs. -1.50 for the rest;
- k-mers containing GATA/TATC (5.3%) are not enriched (-1.48 vs. -1.50);
- the canonical WGATAR core AGATAA/TTATCT occurs in only 182 k-mers, all with count $\leq$9, so none reaches the count$\geq$10 training set.

In the test set, 2 of the top-100 k-mers contain GATA/TATC and 63 contain the ETS core GGAA/TTCC. In the logos, positive relevance falls on the flanking ACTT and GG of ACTTCCGG and on the GGAA run of CCGGAAGT, while the central CC gets little relevance. Measured as the share of a k-mer's one-hot $|R|$ that falls on its GGAA/TTCC core, the values for the top-100 k-mers are 0.342 (DFN (a)), 0.426 (CNN1D (a)), 0.447 (CNN1D (d)), 0.495 (DFN (e)) and 0.542 (CNN1D (e)). The positional expectation for the same positions is 0.349, 0.427, 0.433, 0.417 and 0.465. With one-hot input, relevance therefore follows the motif in individual strong binders but is not concentrated on the 4-bp core beyond what the positional profile predicts. In the shape-only maps (e), which are projected from window relevance (Section 2.10), the core share exceeds the positional expectation by 0.08 for both models (DFN 0.495 vs. 0.417, CNN1D 0.542 vs. 0.465): the MD windows that cover GGAA/TTCC are the ones the models use for strong binders.

4 Discussion

Strand asymmetry. The two orientations of the same 10-mer correlate only $r \approx 0.42$ in log affinity, and this does not improve when only high-count pairs are kept (0.425 at count $\geq$30). The asymmetry is therefore systematic, not counting noise. A likely cause is the orientation-dependent expected counts of the Markov background model, possibly combined with position or flank effects in the selection library. Averaging the two strands destroys this signal. Per-strand labels with reverse-complement-grouped CV improved CV R^2 by 0.15 (RF) and 0.17 (CNN1D) over merged labels.

Count threshold. Counts below 10 add many labels whose relative SE is $\sqrt{2/n}$, i.e. $> 45\%$. A threshold of 30 removes most low-affinity k-mers and biases predictions upwards on the shared evaluation rows. A threshold of 10 was the best trade-off. The log transform was marginally better than raw affinity.

Can MD data alone predict affinity? (the aim). Yes, to a large extent, but the result needs a qualification (Section 3.5).

- **How good.** Tetramer MD only gives CNN1D a test R^2 of 0.700 (Pearson 0.838), 95% of one-hot only (0.739) and of MD + one-hot (0.736). Every model with MD only is within 0.03–0.05 of its one-hot value, except Linear/Ridge and RF, which are better with MD (+0.036, +0.027). Tetramer MD is the best MD-only set (hexamer 0.553, tetramer+hexamer 0.672 for CNN1D).
- **How much sequence is in the MD tables.** Nearly all: the 256 tetramer vectors are separable (rank 27, minimum distance 0.75 in z-units), and a linear classifier decodes the two central bases with 100% accuracy and the outer bases with 75.8%. A model can therefore use the table as a label of the tetramer, and a convolutional network whose first layer is a free lookup is expected to do exactly that.

- **Identity code or physical descriptor.** Both. Randomly reassigning the MD rows to the tetramers lowers R^2 by 0.135 for DFN, 0.052 for RF and 0.022 for CNN1D (0.005 for the averaged CNN prediction). The real values therefore contain more than identity. This advantage is large for models that cannot learn a per-tetramer code (DFN, RF) and small for CNN1D, which with a random table already reaches 0.678. For the deep convolutional models the MD tables are mostly a (good) identity code, and MD never improves on sequence. For Linear/Ridge, RF and DFN MD helps compared with one-hot or random codes.
- **Limitations of the control.** Only the tetramer set was permuted, with 2 permutations and 3 models; Linear, SVR, the attention models and the hexamer sets were not run. The control cannot prove that physics is used, since any physical meaning is itself a function of the tetramer. It does show that the real ordering of the values adds measurable information.

Value of the shape features. The one-hot-only set (d), run with the full final protocol, isolates the contribution of shape (Table 11). Adding tetramer shape (a) to one-hot helps exactly the models that lack a positional, sequence-aware inductive bias:

- Linear/Ridge: 0.380 → 0.471/0.470 (+0.09), the largest gain. A shape value is a fixed non-linear function of four bases, which an additive model cannot build from one-hot itself.
- RF: 0.536 → 0.572 (+0.036); its train R^2 also rises (0.745 → 0.884), as the continuous shape covariates offer many more split points.
- SVR: 0.581 → 0.598 (+0.017); DFN: 0.635 → 0.644 (+0.009); Transformer: 0.679 → 0.685 (+0.006).
- CNN1D: 0.739 (one-hot) vs. 0.736 (a); CNNAttention 0.729 vs. 0.725. The differences are within one sd over runs (0.009–0.016), so shape adds nothing for the convolutional models: with 16 one-hot channels per tetramer window they can learn any per-tetramer lookup, including the shape tables, themselves. Averaged-runs R^2 agrees (CNN1D 0.789 vs. 0.788; CNNAttention 0.785 vs. 0.778).

This confirms the ablation (CNN 0.739 vs. 0.738; RF 0.563 vs. 0.599) under the final protocol with 3 seeds. In the ablation, shape alone was informative (CNN 0.692 without one-hot) and `fce` carried more signal than `avg`.

Model comparison and overfitting.

- Linear and ridge regression underfit (train R^2 0.50, gap 0.03 with set (a); 0.40 and 0.02 with one-hot only).
- Random forest memorised the noisy labels (train 0.884, gap 0.31 with set (a)).
- SVR generalised better (gap 0.15) and was the best classical model (0.598).
- The convolutional models were best on every feature set. Their validation loss reached its minimum early (best epoch ≈22–27) while the training loss kept falling (final train MSE 0.05–0.07 vs. best validation MSE 0.26–0.27). This is overfitting, controlled by early stopping: validation and test R^2 agree closely (CNN1D 0.735 vs. 0.736), so there is no evidence of leakage.
- The Transformer had the smallest gap (0.077–0.089) and nearly equal train and validation MSE (0.293 vs. 0.291). It trained longer (best epoch ≈112) and stayed underfit relative to the CNNs.
- Averaging the runs of one model over folds and seeds helped the NNs substantially (CNN1D 0.736 → 0.788, CNNAttention 0.725 → 0.778, DFN 0.644 → 0.673), and the non-NN models only slightly (+0.002–0.009).

MD-only feature sets (e–g). Tetramer MD (e) is best for every non-linear model, as with one-hot (a vs. b, c). Hexamer MD only (f) is the worst (CNN1D 0.553, DFN 0.529, Linear 0.314), because the 5 hexamer windows leave the outer tetramer windows without information and, without one-hot, nothing compensates. The combination (g) is below (e) for SVR, RF, DFN and all NNs, and above it only for Linear/Ridge (0.443/0.441 vs. 0.416/0.415). In the convolutional models, adding one-hot to MD helps by 0.036 (CNN1D, 0.700 → 0.736), but removing MD from MD + one-hot costs nothing (0.739).

Which shape feature set? Among the shape sets, tetramer (a) gave the highest test R^2 for every non-linear model (Table 11):

- CNN1D: 0.736 vs. 0.714 (hexamer) and 0.723 (both);
- CNNAttention: 0.725 / 0.702 / 0.709;
- Transformer: 0.685 / 0.673 / 0.681;
- DFN: 0.644 / 0.633 / 0.641;
- SVR: 0.598 / 0.586 / 0.588;
- RF: 0.572 / 0.549 / 0.563.

Only the additive models preferred the combined set (Linear 0.484 and Ridge 0.483 vs. 0.471/0.470 with tetramers). Hexamers were the worst shape set for every model; for the CNNs, DFN and Transformer they were even below one-hot only. There are three reasons:

1. Hexamer averages are 96–99% correlated with their central tetramer, so they add little new information.
2. A 10-mer has only 5 hexamer windows against 7 tetramer windows, so the outermost base steps are not described (for windowed models the end positions are zero-padded).
3. The combined set doubles the shape channels (364-dimensional flat input, 70 channels per position) without adding signal, which slightly increases variance for the flexible models.

The additive models gain about 0.013 from the combination because a hexamer value acts as a fixed non-linear (context-dependent) feature of 6 bases. The differences between feature sets (0.01–0.02 for non-linear models) are small compared with the differences between models, but they are consistent across models and comparable to or larger than the sd over runs.

Computational cost. Ridge trains in under 1 s per run. SVR, RF, DFN and the CNNs take about 0.3–1.5 min per run under the parallel load, and the Transformer about 2–5.5 min. Feature building takes under 1 s in all cases. Inference is $\leq$0.3 s per 1,000 sequences for all models except SVR ($\approx$3–4 s with shape, 0.5 s with one-hot only), whose cost scales with the number of support vectors and the input dimension. Dropping shape makes the classical models much cheaper (SVR 58.6 → 10.6 s, RF 42.6 → 2.3 s per run), whereas NN training time barely changes (CNN1D 50.0 vs. 43.6 s). CNN1D on one-hot only therefore gives the best accuracy at moderate cost (Figure 6). MD-only models are cheaper than shape + one-hot for the classical models (SVR 30.5 s and RF 24.2 s per run for (e), vs. 58.6 and 42.6 s for (a)) and similar for the NNs (CNN1D 52.3 s, CNNAttention 62.8 s, Transformer 150.6 s for (e); Table 16). The Transformer hexamer runs (mean 330 s) overlapped with the heaviest parallel load, so part of that difference reflects contention.

Feature importance: which MD parameters, and what do they mean?

- **Which MD parameters.** In the MD-only models, 57–64% of the relevance sits on `fce` off-diagonal couplings, 18% on `fce` diagonal (stiffness of single parameters) and 18–25% on `avg` (mean geometry). Within `avg`, TILT is the most used parameter in both models (3.9% DFN (e), 5.8% CNN1D (e)); the largest couplings are TILT–ROLL (10.6% DFN), SLIDE–TILT (6.1% CNN1D) and ROLL–TWIST (5.3% CNN1D). RISE has the clearest positive signed effect on strong binders (+0.216 DFN (e), +0.120 CNN1D (e)). The stiffness (covariance) terms are thus used more than the mean geometry, consistent with the data containing a sequence-dependent deformability signal. Roll/twist/slide deformability is a plausible indirect-readout signature for an ETS-type GGAA site, which is recognised with DNA bending.

- **Why this is not proof of physics.** Because every shape value is a deterministic function of the tetramer, LRP credit on a parameter can reflect that this parameter helps to tell tetramers apart, not that the TF senses it. The permuted-table control (Section 3.5) is the more direct test. It shows that the real values help beyond identity for DFN (+0.135) and RF (+0.052), and only slightly for CNN1D (+0.022), so the LRP ranking of parameters is most credible for the DFN. The six `avg` parameters and the 21 `fce` terms per window are also correlated with each other, and LRP shares credit between correlated channels.

- **No GATA signal.** LRP does not show relevance concentrated on a GATA core, for a simple reason: `4.txt` contains essentially no GATA signal at count $\geq$10. Its strongest k-mers are ETS-class sites (ACTTCCGG/CCGGAAGT, GGAA core), and the models' relevance logos highlight exactly these bases, also for the MD-only models (core share 0.495 and 0.542 vs. a positional expectation of 0.417 and 0.465). Either the HT-SELEX library selected an ETS-like binder or the file is mislabelled or contaminated. This should be verified against the original SELEX run before any Gata4-specific interpretation.

- **Shape vs. sequence with one-hot.** CNN1D uses one-hot most, yet it still gives about half of its relevance to shape channels in set (a). Because the shape values are a deterministic function of the 16 one-hot channels of the same window, the network can spread the same information over both, and LRP shares the credit between them. This is consistent with the identical accuracy of sets (a) and (d). Without one-hot (e), the shape block takes over the whole signal, and its relevance concentrates on the `fce` couplings.

- **Caveats.** LRP maps for the CNN depend on the rule choice: γ maps correlate only 0.14–0.17 with gradient$\times$input, and only about 50% of the output is conserved. Feature-type shares are partly driven by the number of channels per group. The positional profiles mainly reflect architecture (padding for CNN1D, window coverage for the DFN), and the base-level maps of (e) are projections from window relevance.

SELEX_test.txt mismatch. Only 8.4% of the 5,511 10-mers in `SELEX_test.txt` appear in `4.txt`. On the 462 shared k-mers its values are uncorrelated with the `4.txt` affinities (Pearson of logs -0.03, Spearman -0.06). Its scale is different (max 1.0, weights 1 or 10). It probably comes from another experiment or processing, so it was not used as a test set.

Limitations. The ablation used a single seed and one-factor-at-a-time changes. We did not test de-duplication of shifted motif overlaps, or similarity-based grouping beyond reverse complements, so overlapping k-mers sharing a core motif can still occur on both sides of a split. Hyperparameters were not tuned per model. The only test data come from the same SELEX experiment.

5 Conclusion

Answer to the aim. Good affinity prediction *is* possible from MD simulation data alone: tetramer MD shape and stiffness tables, with no sequence input, give CNN1D a held-out test $R^2 = 0.700 \pm 0.009$ (Pearson 0.838, Spearman 0.821; 0.751 averaged over runs), compared with 0.739 for one-hot only and 0.736 for MD + one-hot. MD-only is thus within 0.04 R^2 (95%) of the sequence-based result, and better than one-hot for Linear/Ridge and RF. Hexamer tables alone are clearly worse (0.553), and combining tetramer and hexamer MD does not help the NNs (0.672). The MD tables identify the tetramer (middle bases decoded with 100%, outer bases with 76% accuracy), and replacing the real values by a random lookup of the same size costs 0.135 R^2 for DFN, 0.052 for RF and only 0.022 for CNN1D. So the MD tables carry real information beyond identity, which models without their own sequence code exploit, whereas the best (CNN) models extract mostly sequence identity from them. LRP on the MD-only models points at `fce` coupling terms (57–64% of relevance; TILT–ROLL, SLIDE–TILT and ROLL–TWIST among the largest) and RISE as the most positive parameter for strong binders, with the caveat that credit on correlated, sequence-determined features is not proof of physical readout.

The new MD shape tables are complete, internally consistent and differ measurably from the earlier tetramer tables. For Gata4 HT-SELEX the optimal preprocessing is log affinity, count $\geq$10, per-strand labels, tetramer avg+fce plus one-hot, and standardisation, evaluated with reverse-complement-grouped CV. With this setup CNN1D reaches a held-out test R^2 of 0.736 (0.788 when its runs are averaged) and is the best of the eight models, ahead of CNNAttention (0.725), the Transformer (0.685), DFN (0.644), SVR (0.598), RF (0.572) and Linear/Ridge (0.47). The feature-set benchmark (a–d) shows:

- Shape features are redundant for the convolutional networks: one-hot only gives the same accuracy (CNN1D 0.739, CNNAttention 0.729) at slightly lower cost.
- Shape features clearly help models without a positional sequence bias: Linear/Ridge +0.09, RF +0.036, SVR +0.017, DFN +0.009 over one-hot only.
- Tetramer tables are the best shape set for every non-linear model; hexamers alone are consistently worst, and combining them with tetramers helps only the linear models (+0.013), consistent with their near-redundancy with the central tetramer.

CNN1D is the best accuracy/cost trade-off (about 45–50 s per run on one CPU thread). LRP on DFN and CNN1D shows that the networks use one-hot and the `fce` coupling terms most, and that their relevance on the strongest binders falls on an ETS-like ACTTCCGG/CCGGAAGT motif. There is no GATA core: `4.txt` has no GATA enrichment at count $\geq$10, which calls the identity of the selected protein into question. The main open issues are this, the strong strand asymmetry of the SELEX labels and the lack of an independent test set consistent with `4.txt`.

A Per-feature-set diagnostic plots

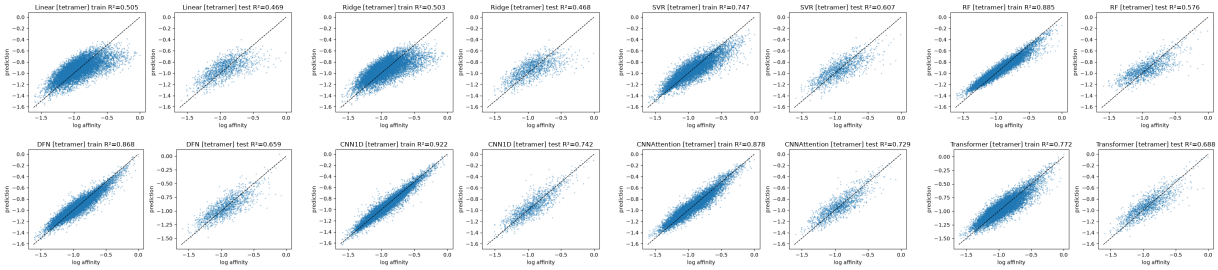


Figure 13: Ground truth vs. prediction (fold 0, seed 0; left half of each panel: train, right half: test) for the tetramer shape + one-hot feature set.

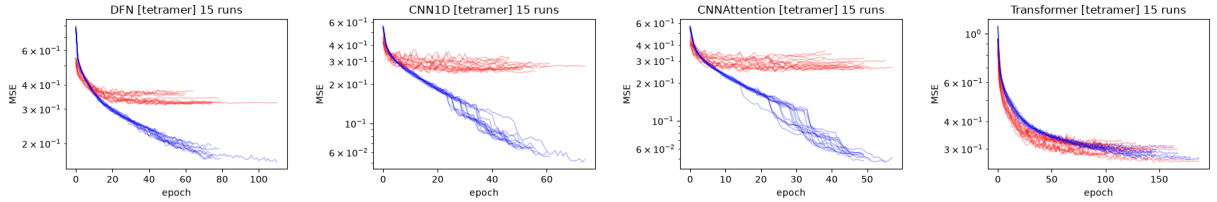


Figure 14: Learning curves (blue: train, red: early-stopping validation; 15 runs each) for the tetramer shape + one-hot feature set.

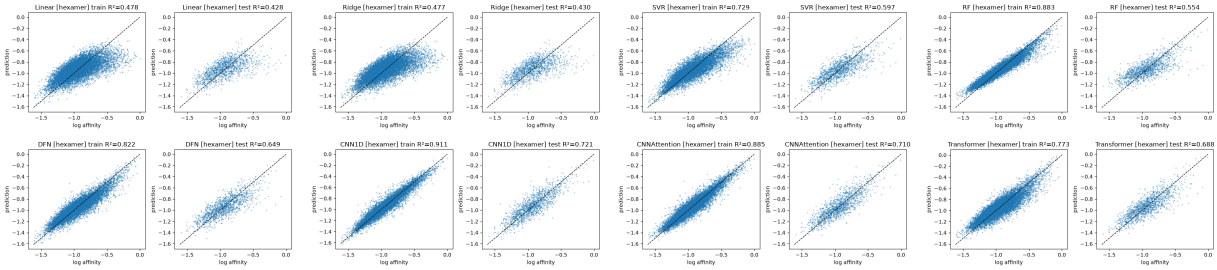


Figure 15: Ground truth vs. prediction (fold 0, seed 0; left half of each panel: train, right half: test) for the hexamer shape + one-hot feature set.

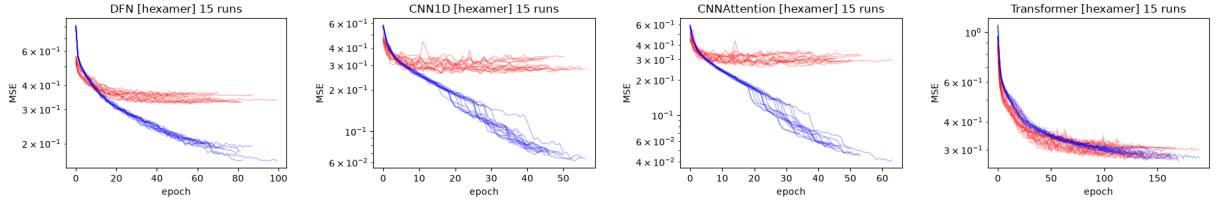


Figure 16: Learning curves (blue: train, red: early-stopping validation; 15 runs each) for the hexamer shape + one-hot feature set.

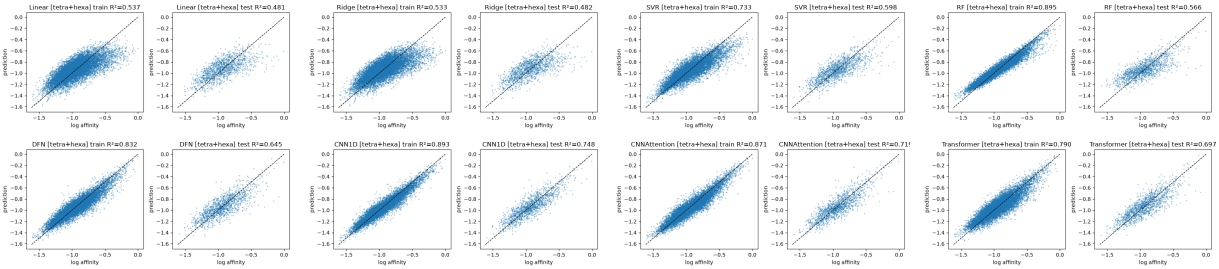


Figure 17: Ground truth vs. prediction (fold 0, seed 0; left half of each panel: train, right half: test) for the tetramer+hexamer shape + one-hot feature set.

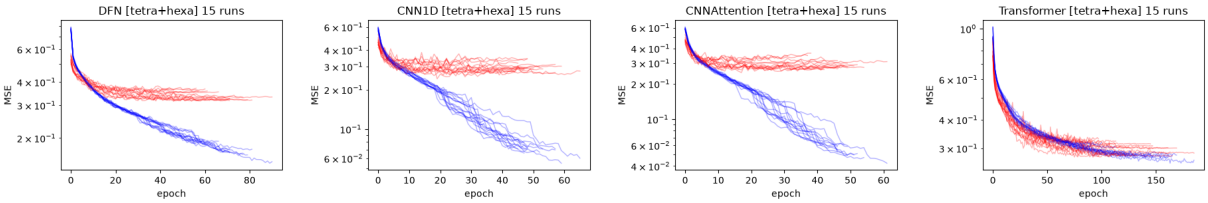


Figure 18: Learning curves (blue: train, red: early-stopping validation; 15 runs each) for the tetramer+hexamer shape + one-hot feature set.

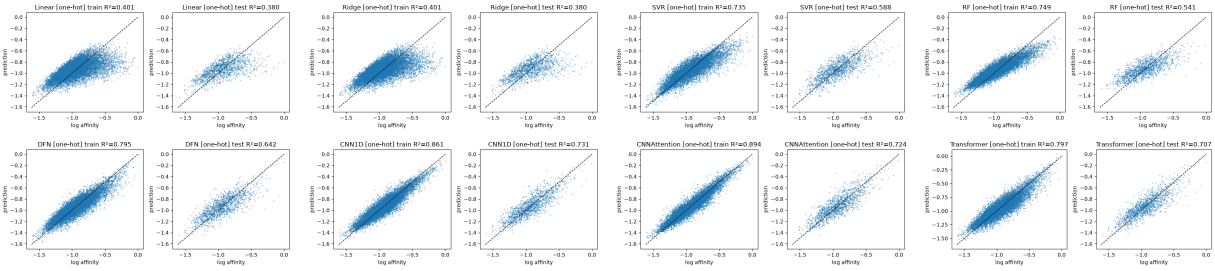


Figure 19: Ground truth vs. prediction (fold 0, seed 0; left half of each panel: train, right half: test) for the one-hot only feature set.

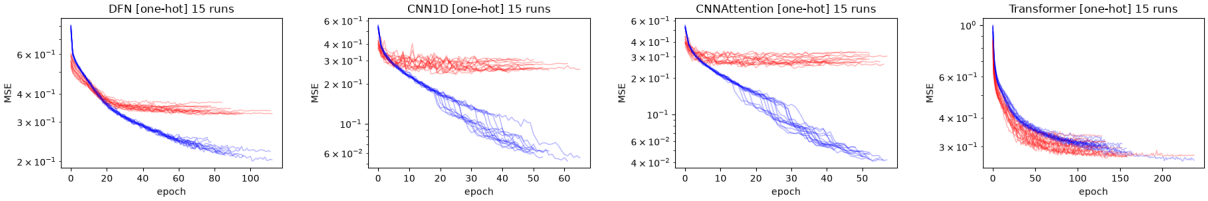


Figure 20: Learning curves (blue: train, red: early-stopping validation; 15 runs each) for the one-hot only feature set.

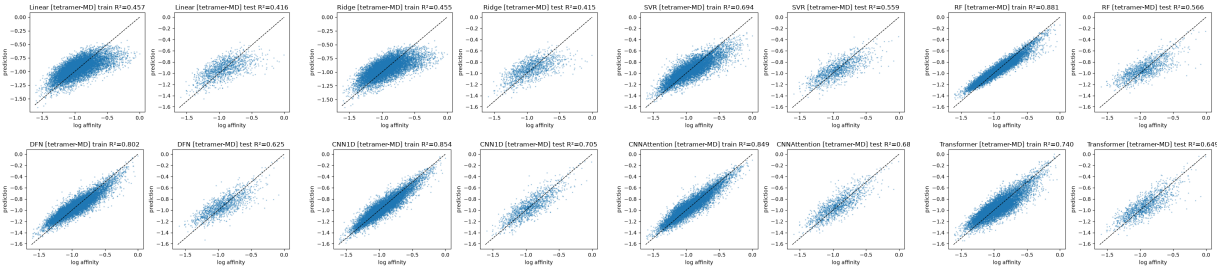


Figure 21: Ground truth vs. prediction (fold 0, seed 0; left half of each panel: train, right half: test) for the tetramer MD-only (set e) feature set.

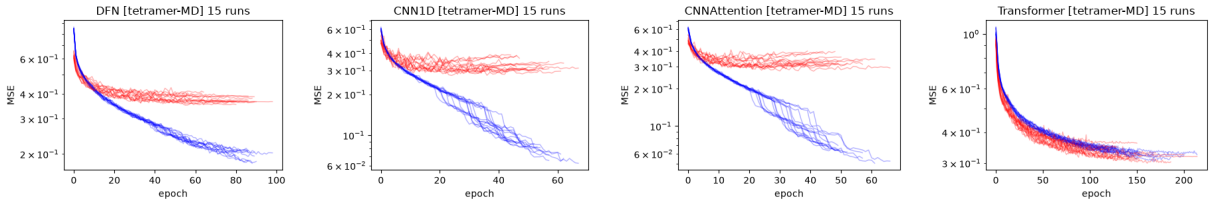


Figure 22: Learning curves (blue: train, red: early-stopping validation; 15 runs each) for the tetramer MD-only (set e) feature set.

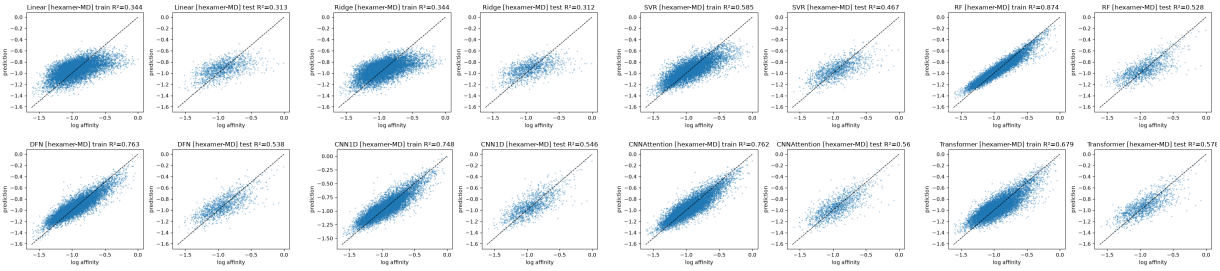


Figure 23: Ground truth vs. prediction (fold 0, seed 0; left half of each panel: train, right half: test) for the hexamer MD-only (set f) feature set.

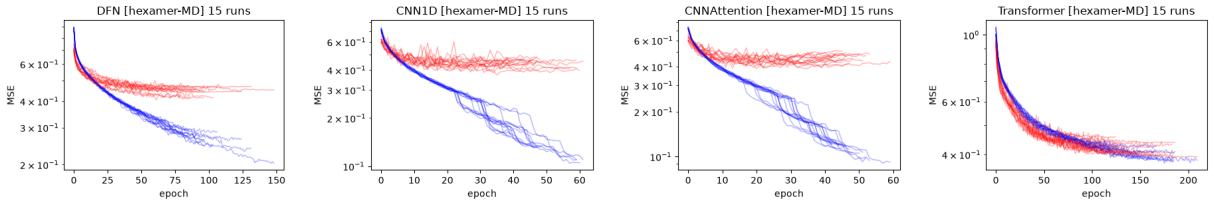


Figure 24: Learning curves (blue: train, red: early-stopping validation; 15 runs each) for the hexamer MD-only (set f) feature set.

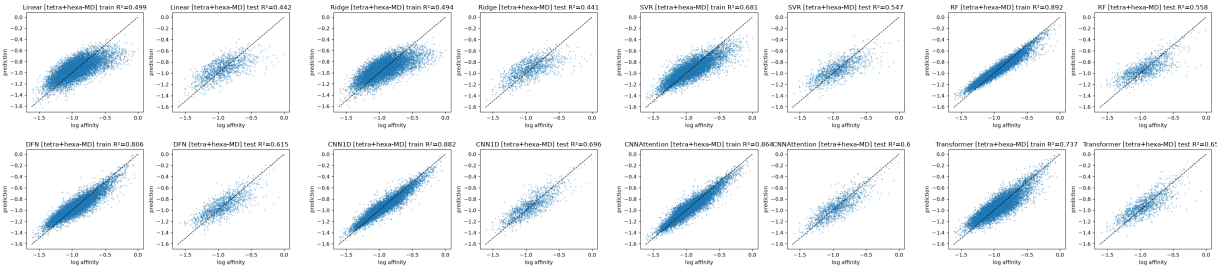


Figure 25: Ground truth vs. prediction (fold 0, seed 0; left half of each panel: train, right half: test) for the tetramer+hexamer MD-only (set g) feature set.

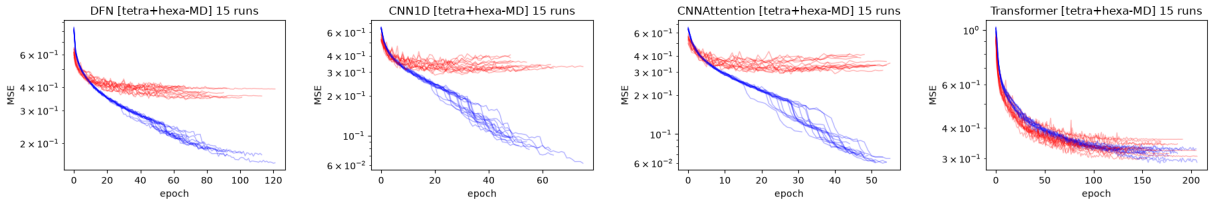


Figure 26: Learning curves (blue: train, red: early-stopping validation; 15 runs each) for the tetramer+hexamer MD-only (set g) feature set.

B Permuted-table control plots

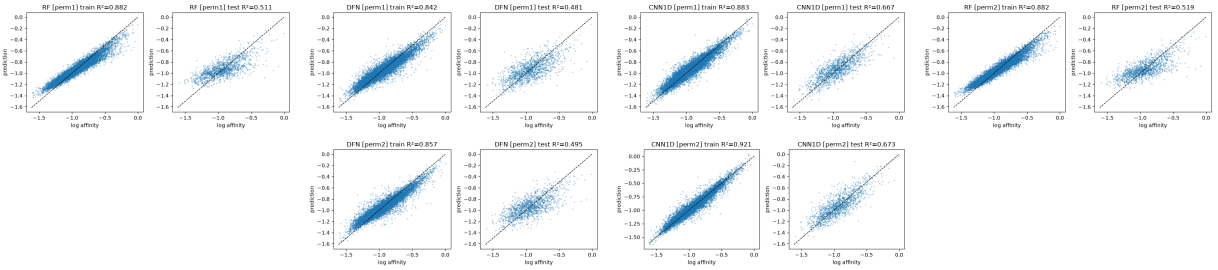


Figure 27: Ground truth vs. prediction (fold 0, seed 0) for RF, DFN and CNN1D on set (e) with the MD rows randomly reassigned to tetramers (first row: permutation 1; second row: permutation 2).

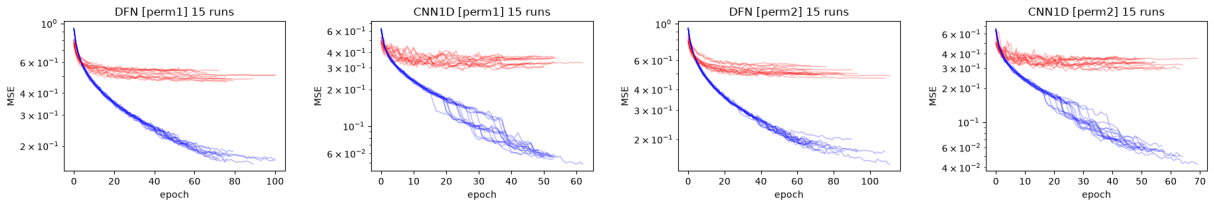


Figure 28: Learning curves for DFN and CNN1D with permuted MD tables (permutation 1: first two panels; permutation 2: last two).

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